

Griffith School of Engineering and Built Environment
Griffith University

3004ENG — Project Management Principles

Trimester 2, 2026

Eris TestFlight2 Launch Campaign

Project Execution Plan

Assessment 3: Group Project Execution Plan — Group 1

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Acting as a project management consultancy engaged to deliver launch-site operations at Bowen Orbital Spaceport.

Contribution

Table 1: Agreed contribution after hours equalisation. Each member 42 h; percentages match the title page and sum to 100.0%.

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Hadi Alkhub (s5331680)	42	Scope; schedule; network logic; Gantt; compression	25.0
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We confirm that the contribution percentages above and the submission below are agreed by all members.

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1 Executive summary

This Project Execution Plan asks the Gilmour Executive Board to approve or decline Eris TestFlight2 *launch-site operations* at Bowen Orbital Spaceport from this section alone. TestFlight 1 cleared the tower and ended at T+14 s after thrust loss in two first-stage hybrid motors, from electrical and thermal pump faults (Smith, 2025; Gilmour Space Technologies, 2025; Dougherty, 2026). Group 1 is a project-management consultancy. In-scope work is pad operations: cleanroom receipt, 23 m stacking, cryogenic GSE, a cold Wet Dress Rehearsal, a five-second static fire of the four first-stage hybrid engines, range clearance, lift-off, telemetry handover and closeout. Factory manufacture, engine redesign and freight to the receiving gate stay with Gilmour and sit outside this baseline.

The work estimate is \$1,500,000. Contingency is \$78,890 (5.26% of base): \$35,760 of residual expected monetary value on ten cost-bearing risks plus a \$43,130 itemised in-scope residual-uncertainty allowance. Authorised *BAC* is \$1,578,890. Contingency may be consumed; nothing beyond *BAC* is authorised. Delivery is a managing-contractor / design-and-construct hybrid: Gilmour keeps propulsion intellectual property, vehicle integration and the launch licence; civil and cryogenic GSE packages are let design-and-construct with an interface-control freeze and liquidated damages.

The campaign uses a seven-day pad calendar from 10 August to 1 December 2026, because LOX load, static fire and *T-0* do not pause for weekends. Baseline dates: stacking 15 October 2026 (M-4 / A-123); cold Wet Dress Rehearsal 24 October (A-132); static fire 30 October 2026 (M-5 / A-133); Launch Readiness Review 10 November 2026 (M-6 / A-141); *T-0* **15 November 2026** (M-7 / A-143), inside the charter's Q1 2027 window; telemetry handover 20 November 2026 (M-8 / A-144), at least 98.5% of packets; closeout 1 December 2026 (M-9 / A-145). A five-day pull-in to 10 November exists only if R-14 is realised, as a \$21,000 draw on the reserve. It is not the baseline.

Three register items can stop *T-0* or close the heritage case: R-01, R-12 and R-14. R-01 is a residual pump-insulation defect: static-fire auto-abort on A-133 and a 12–14 day critical-path recycle. R-12 is the TestFlight 1 in-flight mode: thrust loss in two first-stage hybrid motors after *T-0*, as at T+14 s, ending the heritage objective. R-14 is an ASA flight-authorisation or CASA NOTAM refusing the 15 November slot. Occupancy is held at the pad occupancy cap of twelve after an unmitigated peak of fourteen. The organic crew is eight.

It is recommended that the Executive Board approve this Project Execution Plan, baseline the campaign at \$1,578,890 AUD (inclusive of \$78,890 contingency), including the \$602,301 of planned pad PV already on the clock by Gate 1 on 2 October 2026 (40.2% of the work PMB: permitting, Juru, cleanroom, GSE, RF and stacking in progress), lock remaining spend at that gate, and hold T-0 on 15 November 2026 inside the charter's Q1 2027 outer window.

2 Refined project overview

2.1 Background and purpose

This is the Week 11 reading of a campaign chartered on 7 August 2026. Gilmour Space Technologies flew Eris TestFlight 1 from Bowen in July 2025. The vehicle cleared the tower and was terminated at T+14 s after thrust loss in two first-stage hybrid motors (Smith, 2025; Gilmour Space Technologies, 2025; Australian Space Agency, 2025). TestFlight 2 is the late-2026 follow-on, inside the charter's Q1 2027 outer window (Dougherty, 2026). Group 1 plans and controls launch-site operations, listed in Section 2.2. Manufacturing, engine research, freight to the gate, and pump redesign remain Gilmour factory work. Orbital insertion is Gilmour's vehicle-success test, not a consultancy finish-line metric.

The Bowen facility licence has been held since March 2024. Stacking (A-123) is a crane lift, mate and bolt of the 23 m vehicle, with no ignition, so that licence already covers it. Hot fire (A-133) is the five-second static fire of the four first-stage hybrid engines, vehicle still held down. It waits on Australian Space Agency (ASA) flight authorisation, Civil Aviation Safety Authority (CASA) airspace, and Great Barrier Reef Marine Park Authority (GBRMPA) marine-park permission (Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019; Australian Space Agency, 2025). Those instruments gate hot fire, not stacking. ASA stays Keep Satisfied, filed 60 days prior on A-113. The CASA Notice to Airmen (NOTAM) is the A1 M-3 airspace instrument and does not replace ASA. The Juru people stay High power / High interest, Manage Closely, now with stop-work authority before Wet Dress Rehearsal. The payload sponsor stays Keep Informed, witnesses LRR on A-141, and has no WBS package.

2.2 Scope

Scope is on the site operations.

Included:

- WBS 1.1–1.4: campaign governance and EVM (1.1), pad logistics and GSE (1.2), pad testing (1.3), and flight, launch and closeout (1.4).
- Seventeen Level-3 packages in Figure 2.1; those are the cost accounts.
- Two extra schedule bars in Table 3.1 (nineteen bars in total): the split of package 1.2.1 into cleanroom (A-121) and unpacking (A-122), and the external vehicle-receipt wait (A-100).

Excluded:

- Factory manufacture.
- Engine R&D.
- Line-haul to the gate.

- Payload operations.
- Any unidentified Level-3 box that would mix factory product into a site baseline.

A-100 is a zero-dollar external driver that pins A-122; it is not a consultancy-controlled launch-critical bar. It is omitted from Table 4.2. Design defects found on the pad are packaged back to R&D. Only a pre-kitted spare may be swapped on A-134.

2.3 Charter baseline versus PEP refinement

This is not a copy of the Week 4 charter. Table 2.1 lists every main change and why it was made. Two Assessment 1 dates did not match: stacking complete (M-4) is now A-123 on 15 October 2026, and telemetry handover (M-8) is now A-144 on 20 November. Short forms in that table: LRR (Launch Readiness Review), LEO (Low Earth Orbit), HSE (Health, Safety and Environment), EMV (expected monetary value), RES (residual-uncertainty allowance), WDR (Wet Dress Rehearsal), LOX (liquid oxygen).

Table 2.1: Charter baseline versus PEP refinement.

Project element	A1 charter (Week 4)	A3 PEP (Week 11)	Why it changed
M-4 stacking date	Objective 1: pad integration 15 Oct. Milestone M-4: stacking 28 Oct.	M-4 locked to 15 Oct 2026 (A-123 <i>EF</i>).	Removes the internal collision. Creates a 15-day pad-test buffer before the 30 Oct hot-fire gate.
M-8 handover date	“5 days post-launch” and milestone date 22 Nov (7 days after 15 Nov).	M-8 = 20 Nov 2026 (A-144).	Aligns the milestone with Objective 5 and the 5-day rule.
Cost baseline	“Manage within the allocated operational budget.” No numbers figure.	Bottom-up \$1,500,000 + \$78,890 contingency (residual EMV \$35,760 + RES \$43,130).	Crane hire, cryogenic crews, and LOX delivery cannot be controlled without a numbered baseline.
Permit vs stacking logic	M-3 flight permits 25 Oct were predecessors of M-4 stacking (then 28 Oct).	Facility licence already held (Mar 2024). A-113 gates A-133 static fire, not stacking. Permits <i>EF</i> 24 Oct; hot fire <i>ES</i> 25 Oct.	Stacking is a site-licence activity. Hot fire is a range activity. This also creates a second critical path.
Juru engagement	High power / high interest, Manage Closely, consultation and monitoring.	Same power/interest (High / High, Manage Closely). Embedded monitors with stop-work authority and a cultural hold point before WDR (A-114 → A-132).	A1 strategy was correct in class; the PEP makes it an enforceable pad control, not a comms plan.

Table 2.1: Charter baseline versus PEP refinement (continued).

Project element	A1 charter (Week 4)	A3 PEP (Week 11)	Why it changed
Testing sequence	WDR and 5-second static fire as one Objective 3 output by 30 Oct.	A-132 WDR (cold) 16 Oct–24 Oct, then A-133 static fire 25 Oct–30 Oct. LRR waits on A-134 diagnostics.	Hot fire must wait for flight permits. LRR cannot start on incomplete post-fire data.
Risk governance	Five high-level threats, no $P \times I$ money.	18 campaign-specific risks, Cause / Event / Effect columns, EMV \$35,760 + RES \$43,130.	Generic rows cannot size contingency or point at a WBS line.
Resource peak	Roles listed; no demand vs capacity.	Organic crew 8 named heads (GSE1, GSE2, GSE3, GSE4, STK1, STK2, LD, RSO), pad HSE cap 12 , unmitigated peak 14 , mitigated peak 12 .	A1 could not show where the plan exceeds capacity.
Regulator and payload	ASA High/Low Keep Satisfied, file 60 days prior. Payload sponsor Med/High Keep Informed, witness LRR.	ASA remains Keep Satisfied and is filed 60 days prior on A-113. CASA issues the NOTAM as the instrument of A1 M-3 civil aviation airspace arrangements, not as a substitute for ASA. Payload sponsor still witnesses LRR on A-141 (HP-4). No 10–12 Nov geometry is used.	A1 named ASA, not Townsville, as the high-power regulator. The PEP keeps that filing rule and uses CASA only as the airspace instrument.
Success criteria	Q1 2027 window; no funds beyond contingency; 98.5% telemetry in 5 days; zero LTI/GBRMPA/Juru; defects to R&D.	T-0 15 Nov inside A1's Q1 2027 window (2-day in-flight trigger); $AC \leq BAC$; 98.5% telemetry in 5 days; LRR first submission; four-engine 5 s fire; zero LTI/GBRMPA/Juru; defects to R&D. LEO is Gilmour's test, not the consultancy's.	A1 tests stay. The PEP numbers cost and tightens in-flight control inside Q1 2027. LEO is not this team's finish line.
Objectives 4 and 5	Objective 4: pass LRR by 10 Nov. Objective 5: November 2026 window and 98.5% telemetry.	Objective 4 is LRR by 10 Nov (M-6 / HP-4, first submission). Objective 5 is the November window (T-0 15 Nov) plus 98.5% handover on 20 Nov. Closeout on 1 Dec is a success test, not a numbered objective.	A3 keeps A1's numbered Obj 4 as LRR rather than remapping it onto T-0.

Table 2.1: Charter baseline versus PEP refinement (continued).

Project element	A1 charter (Week 4)	A3 PEP (Week 11)	Why it changed
LRR attendees	M-6: Launch Director, Range Safety Officer and Juru Land Liaisons review. Payload sponsor witnesses LRR.	HP-4 release is Launch Director, RSO, Juru Officer, Abdul and Ziyad. Payload sponsor remains a witness, not a signatory.	A1 named the review board. The PEP keeps those three on the hold and does not promote the payload sponsor from witness to signatory.
Community and media stakeholders	Bowen Community Keep Informed: clearance maps 14 days prior. QLD Regional Press Monitor: post-launch press releases.	Community notice 14 days prior (R-17). No named media stakeholder. Clearance maps are not a PEP artefact.	The 14-day community notice A1 required stays as an HSE control. A Monitor-only press party is not a pad control.
WDR T-10 acceptance	M-5 acceptance: countdown to 10 seconds before simulated lift-off, plus a 5-second static fire with zero critical anomalies.	A-132 is cold WDR (16–24 Oct). A-133 is the five-second static fire (25–30 Oct). The T-10 simulated-lift-off criterion is not restated.	Cold WDR is LOX load and countdown rehearsal without ignition. T-10 simulated lift-off is not a separate success test in this PEP.
Ilsa Hashmi project role	Project role Stakeholder & Resource Lead. A3 column already Cost, Resource, Capacity Planning.	Cover: Cost, Resource & EVM Lead. Organogram: Cost and Resource Lead. Stakeholder Lead is not used.	A1's A3 column already pointed at cost and resource. EVM is the A3 add. The Stakeholder Lead title is not on the PEP cover or organogram.
Field spend before Gate 1	M-1 charter 7 Aug. M-2 / Gate 1 PEP lock 2 Oct. Phase 3 execution follows M-2.	Charter has no NTP. Permitting (A-113) and Juru (A-114) start 10 Aug. Pad occupancy before Gate 1 is a PEP choice. Gate 1 PV \$602,301.	Permit and heritage clocks cannot wait until 2 Oct and still finish before hot fire. Remaining spend still locks at Gate 1.

2.4 Objectives and success criteria

The Week 4 objectives still stand. The network and the budget locked the dates and the dollars. Each test is checked in Sections 3–7.

Objective 1 stayed stacking. A1 also put M-4 on 28 October, so one job had two dates and no cold-flow window before hot fire. M-4 is now A-123 on 15 October 2026.

Objective 2 was only “manage within the allocated budget.” Crane hire, cryogenic crews and LOX need a figure. The PEP uses a \$1,500,000 base and \$78,890 of residual-EMV-plus-RES contingency, audited at M-4, M-5 and M-7.

Objective 3 was one 30 October output: WDR and the five-second fire together. Hot fire waits for flight permits, so A-132 is cold WDR (16–24 October) and A-133 is the five-second static fire (25–30 October).

Objective 4 stayed pass LRR by 10 November (A-141 / M-6 / HP-4), on the first formal submission. That date was not moved onto lift-off.

Objective 5 stayed the November window and 98.5% telemetry. A1's M-8 was 22 November, two days past the five-day rule. Lift-off is 15 November (A-143 / M-7) inside the Q1 2027 window, with a more-than-two-day in-flight slip trigger, and handover is 20 November (A-144 / M-8). Closeout on 1 December (A-145 / M-9) is a success test, not a new Objective 6.

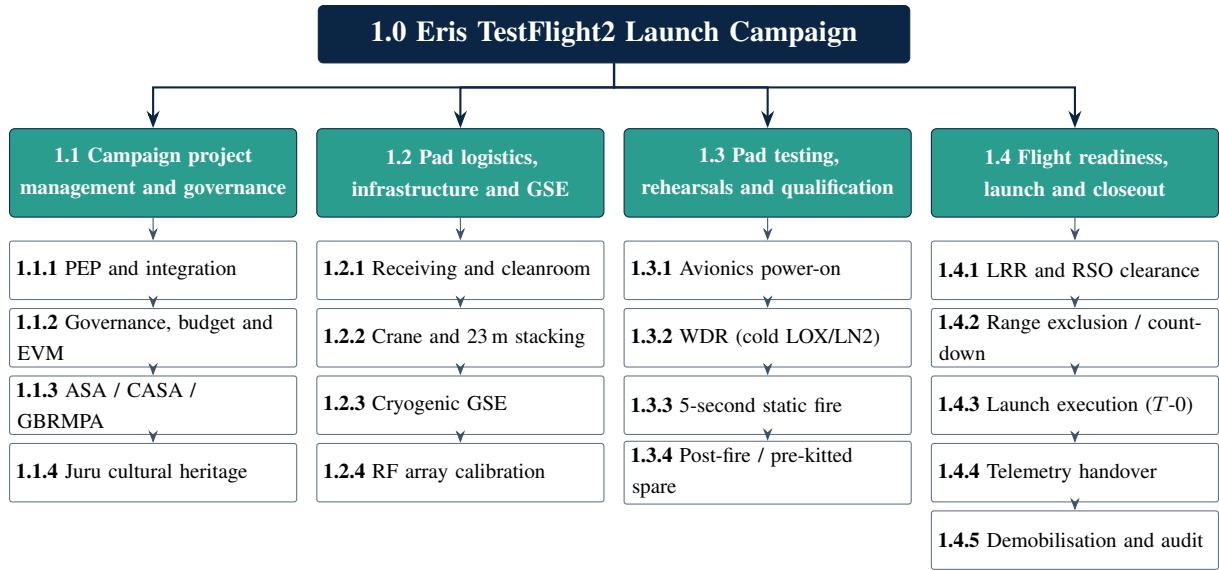
Finish-line success, as used in earned value, is final $AC \leq \$1,578,890$ and $T-0$ on 15 November with that > 2 day trigger. Contingency may be used; nothing above BAC is authorised. HSE stays as chartered: zero lost-time injuries, zero GBRMPA breaches, and zero Juru heritage breaches. Design defects go back to R&D. Reaching LEO is Gilmour's test, not this consultancy's.

2.5 Work breakdown structure

Figure 2.1 decomposes launch-site operations into four Level-2 workstreams and seventeen Level-3 cost accounts. Every scheduled activity except A-100 maps to one package, and every dollar in Table 4.2 sits in one of them. A-100 is the external zero-dollar freight wait that drives A-122. It stays on the network and off the cost accounts.

Package 1.2.1 is one cost account with two bars (A-121 cleanroom and A-122 receiving) because receiving needs both the cleanroom and the vehicle. The other sixteen Level-3 codes match Table 3.1 one-to-one. Factory manufacture, engine research and development, and line-haul to the gate are excluded.

The 7 August charter has no notice to proceed for field work. Permitting (A-113) and Juru mobilisation (A-114) start on 10 August so those clocks finish before hot fire. Pad occupancy before Gate 1 (A-111, 2 October) is a PEP choice. Gate 1 locks the cost and schedule baseline.



EXT A-100 — not a WBS cost account. External \$0 freight driver into A-122; omitted from Table 4.2.

Figure 2.1: Work breakdown structure for Eris TestFlight2 launch-site operations. Seventeen Level-3 packages are the cost accounts in Table 4.2. A-100 is an external schedule bar, not a seventeenth-plus cost line.

3 Schedule management

The campaign schedule turns the WBS into a dated baseline. Dates come from a finish-to-start activity-on-node network on a seven-day pad calendar, 10 August to 1 December 2026, because LOX load, static fire and lift-off do not stop for weekends. Baseline $T-0$ is 15 November 2026, inside the charter’s Q1 2027 window. A five-day pull-in to 10 November exists only if R-14 is realised. It is not the baseline.

3.1 Network logic and float analysis

Links are finish-to-start with zero lag, drawn activity-on-node. Textbook discrete-time notation writes $EF = ES + D$ when EF is exclusive. Every date on the Bowen calendar is inclusive. The identities used here are

$$ES_i = \max_{j \in P_i}(EF_j) + 1, \quad EF_i = ES_i + D_i - 1, \quad (1)$$

$$LF_i = \min_{k \in S_i}(LS_k) - 1, \quad LS_i = LF_i - D_i + 1, \quad (2)$$

P_i and S_i are the predecessor and successor sets of activity i . Where P_i is empty, ES is the campaign start. Total float and free float are

$$TF_i = LS_i - ES_i = LF_i - EF_i, \quad FF_i = \min_{k \in S_i}(ES_k) - EF_i - 1, \quad (3)$$

If S_i is empty, $FF_i = TF_i$. Launch-critical means $TF_i = 0$ on a path that sets $T-0$ or closeout. Table 3.1 is the full pass. It matches the locked register.

The forward pass gives two $TF = 0$ chains. They stay independent until hot fire.

- Vehicle path: A-100 $\rightarrow$ A-122 $\rightarrow$ A-123 $\rightarrow$ A-132 $\rightarrow$ A-133. A-100 is the external \$0 freight wait, not a Gilmour launch-critical bar.
- Permit path: ASA / CASA / GBRMPA permits (A-113) into static fire (A-133).

RF calibration (A-125) feeds static fire, so the array is live before hot fire. It is not launch-critical. After static fire, A-134 through closeout (A-145) is one critical chain. The PEP lock (A-111) finishes on 2 October by an imposed Must-Finish-On (Gate 1). It is not launch-critical. The Week 4 charter approved the document, not field mobilisation.

The paths meet in two staged merges.

- Wet Dress Rehearsal waits on stacking, cryogenic GSE (A-124), avionics power-on (A-131) and Juru monitors (A-114). That is the technical merge.
- Static fire then waits on the rehearsal, flight permits *and* RF calibration. That is the regulatory and range-instrument merge.

Launch Readiness Review (LRR, A-141) waits on post-fire diagnostics, not on static fire. Signing that flight-clearance gate on incomplete hot-fire data would break the hold-point logic of an AS9100D system (SAE International, 2016).

Feeder float is not interchangeable.

- Avionics has one day into the rehearsal.
- Cryogenic GSE has three days.
- RF calibration has 39 days. It sits in August–September so it does not sit on the October pad peak. Moving it does not change 15 November.
- Cleanroom mobilisation (A-121) has $TF = 3$ but $FF = 0$, so delay is absorbed before $T-0$ only by consuming the GSE float.

Total float protects $T-0$. Free float protects the successor. Figure 3.1 draws the links.

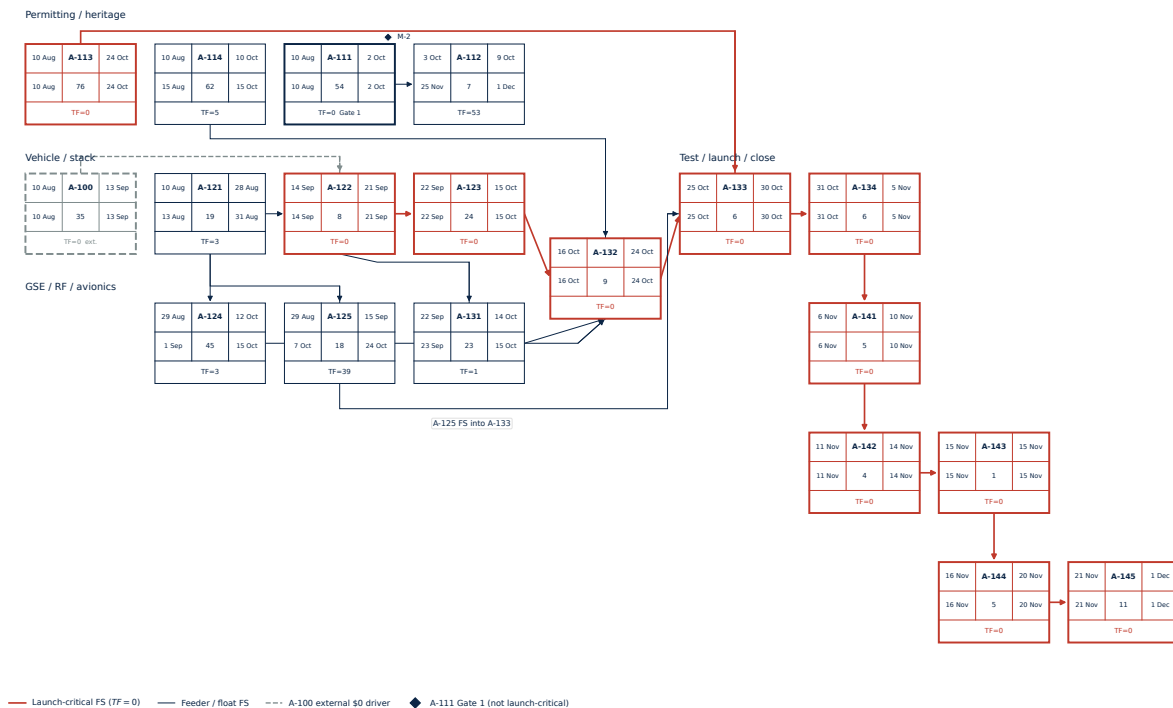


Figure 3.1: Activity-on-node network for the Bowen campaign. Read each box top to bottom: earliest start, ID, earliest finish; latest start, duration in days, latest finish; total float. When the two date rows match, float is zero. Crimson arrows are launch-critical finish-to-start links. The vehicle path and the permit path merge at static fire; RF calibration is the labelled south-routed feeder into the same merge; LRR waits on post-fire diagnostics, not on static fire. The PEP lock is Gate-1 constrained and is not launch-critical. Freight receipt is a grey dashed external \$0 driver, not a crimson Gilmour bar.

Table 3.1: Master campaign activity precedence and float register. Dates are inclusive. $FF = \min(ES_{\text{succ}}) - EF - 1$ days, or $FF = TF$ if there is no successor. The PEP lock $TF = 0$ is an imposed Gate 1 finish, not launch-critical. The freight wait $TF = 0$ is an external pin (CP = Ext), not a Gilmour launch-critical bar.

ID	WBS	Activity	D	Pred.	ES	EF	LS	LF	TF	FF	CP
A-100	EXT	Vehicle receipt (external driver)	35	—	10 Aug	13 Sep	10 Aug	13 Sep	0	0	Ext
A-111	1.1.1	PEP and integration baseline	54	—	10 Aug	2 Oct	10 Aug	2 Oct	0*	0	Gate
A-112	1.1.2	EVM control stand-up	7	A-111	3 Oct	9 Oct	25 Nov	1 Dec	53	53	N
A-113	1.1.3	ASA / CASA / GBRMPA permits	76	—	10 Aug	24 Oct	10 Aug	24 Oct	0	0	Y
A-114	1.1.4	Juru monitor mobilisation	62	—	10 Aug	10 Oct	15 Aug	15 Oct	5	5	N
A-121	1.2.1a	Site mobilisation and cleanroom	19	—	10 Aug	28 Aug	13 Aug	31 Aug	3	0	N
A-122	1.2.1b	Receiving and unpacking	8	A-100, A-121	14 Sep	21 Sep	14 Sep	21 Sep	0	0	Y
A-123	1.2.2	Crane and vertical stacking	24	A-122	22 Sep	15 Oct	22 Sep	15 Oct	0	0	Y
A-124	1.2.3	Cryo GSE piping and checkouts	45	A-121	29 Aug	12 Oct	1 Sep	15 Oct	3	3	N
A-125	1.2.4	RF telemetry calibration	18	A-121	29 Aug	15 Sep	7 Oct	24 Oct	39	39	N
A-131	1.3.1	Avionics power-on sweeps	23	A-122	22 Sep	14 Oct	23 Sep	15 Oct	1	1	N
A-132	1.3.2	Wet Dress Rehearsal (cold)	9	A-114, A-123, A-124, A-131	16 Oct	24 Oct	16 Oct	24 Oct	0	0	Y
A-133	1.3.3	5-second static fire	6	A-132, A-113, A-125	25 Oct	30 Oct	25 Oct	30 Oct	0	0	Y
A-134	1.3.4	Post-fire / pre-kitted spare	6	A-133	31 Oct	5 Nov	31 Oct	5 Nov	0	0	Y
A-141	1.4.1	LRR and RSO clearance	5	A-134	6 Nov	10 Nov	6 Nov	10 Nov	0	0	Y
A-142	1.4.2	Range exclusion and countdown	4	A-141	11 Nov	14 Nov	11 Nov	14 Nov	0	0	Y
A-143	1.4.3	Launch execution ($T-0$)	1	A-142	15 Nov	15 Nov	15 Nov	15 Nov	0	0	Y
A-144	1.4.4	Telemetry decrypt and handover	5	A-143	16 Nov	20 Nov	16 Nov	20 Nov	0	0	Y

Table 3.1: Master campaign activity precedence and float register (continued).

ID	WBS	Activity	<i>D</i>	Pred.	ES	EF	LS	LF	TF	FF	CP
A-145	1.4.5	Demobilisation and closeout	11	A-144	21 Nov	1 Dec	21 Nov	1 Dec	0	0	Y

3.2 Schedule baseline, Gantt visualisation and milestone integrity

The dates in Table 3.1 are the schedule baseline. Figure 3.2 plots the same ES/EF values. August and September carry permitting, heritage, mobilisation and GSE. October is the physical peak: stacking 15 October, WDR 16–24 October, static fire 25–30 October. November is the flight-clearance tail.

Two Week 4 collisions are closed here. Objective 1 dated pad integration 15 October and M-4 stacking 28 October. M-4 is now stacking on **15 October 2026**, which restores the cold-flow window before hot fire. “Within five days of lift-off” was also dated M-8 at 22 November, seven days after a 15 November $T-0$. M-8 is now **20 November**. The facility licence is already held, so permits gate static fire, not stacking.

Three gates sit on the baseline.

- Gate 1 is the PEP lock on 2 October.
- Gate 2 is permits on 24 October. Without those instruments static fire cannot start. That path is already $TF = 0$.
- Gate 3 is LRR on 10 November.

Crashing stacking without crashing the permit path therefore cannot pull $T-0$.



Figure 3.2: Baseline Gantt from 10 August to 1 December 2026, plotted from the same ES/EF values as Table 3.1. Crimson bars are launch-critical. The hatched grey bar is the external \$0 freight wait. Teal whiskers show total float to LF . Vertical markers are M-2 (navy, Gate 1) and M-4 through M-9. Option C is not drawn on this baseline.

3.3 Schedule compression: crashing and fast-tracking

Fast-tracking overlaps sequential work, usually at zero direct cost. Crashing adds resource on critical work and is paid as a cost slope. The Board question is whether $T-0$ can move from 15 November to 10 November if R-14 is realised. Five days must come off a $TF = 0$ path that sets lift-off. After the static-fire merge that path is diagnostics through lift-off. Crashing stacking by five days saves **zero** days on $T-0$ while permits still finish on 24 October. Overlapping stacking with cryogenic GSE also saves zero days: GSE already runs in parallel.

The crash-cost slope on a crashed activity is

$$\text{Cost slope} = \frac{\Delta C}{\Delta T} = \frac{\text{crash cost} - \text{normal cost}}{\text{normal duration} - \text{crash duration}}. \quad (4)$$

Table 3.2 and Figure 3.3 compare three options that each claim five days. Only Option C is both CPM-valid and acceptable against the LRR hold.

Table 3.2: Schedule compression evaluation. All three options save five days on $T-0$ if they work as drawn. Baseline $T-0$ stays 15 November. Option C is a draw on the \$78,890 reserve if R-14 fires, not an addition to the \$1,500,000 base.

Option	Activities modified	Days	Added cost	Slope	Secondary risks	Verdict
A Pure fast-track	Diagnostics → LRR SS+1 (review overlaps diagnostics)	5	\$0	\$0/d	Review before diagnostic close-out; LRU rework; AS9100D hold-point failure	Rejected
B Pure crash	Diagnostics crashed 6 → 1 d	5	\$40,000	\$8,000/d	Fatigue; skipped thermal/electrical screens; recycle into LRR	Rejected
C Hybrid (if R-14 fires)	Diagnostics 6 → 1 d extra labour (\$21,000); LRR kept at 5 d; countdown kept at 4 d	5	\$21,000	\$4,200/d	Screens kept; R-15 4 d marine hold intact	Contingent

Decision: baseline $T-0$ stays 15 November. If R-14 fires, the Board draws Option C and lift-off moves to 10 November. That is the only option valid on the network and acceptable on quality.

Option A is a pure fast-track. It buys five days at \$0 by starting LRR one day after diagnostics start, while those diagnostics are still open. HP-4 is a stop: flight clearance cannot start on incomplete post-fire data (SAE International, 2016). Signing early is a hold-point failure. Waiting until diagnostics finish does not save the five days; it creates rework. Option A is rejected.

Option B is a pure crash of diagnostics from six days to one. The slope is $\$40,000/5 = \$8,000$ per day. A 24-hour war room on a vehicle that has just fired is the work most likely to skip thermal and electrical screens. Option B is rejected.

Option C takes the five days only from diagnostics, and extra labour keeps the screens

Option B would skip: two specialists and two technicians for five days ($2 \times \$165 + 2 \times \$95 = \$520/\text{h} \times 8 \text{ h} = \$4,160/\text{d}$, published at the $\$4,200/\text{d}$ crawler rate). The slope is $\$21,000/5 = \$4,200$ per day. LRR stays five days and countdown stays four, so the R-15 marine hold is kept. If invoked: diagnostics 31 October, review 1–5 November, countdown 6–9 November, lift-off **10 November**. The $\$21,000$ comes from the $\$78,890$ contingency only if R-14 is realised, not from the $\$1,500,000$ base. Feeder float stays at 1 day (avionics) and 3 days (cryogenic GSE).

Option C is chosen because it is cheaper than Option B ($\$4,200/\text{d}$, not $\$8,000/\text{d}$), it keeps the LRR hold Option A would break, and crashing stacking cannot move $T-0$ while permits finish on 24 October.

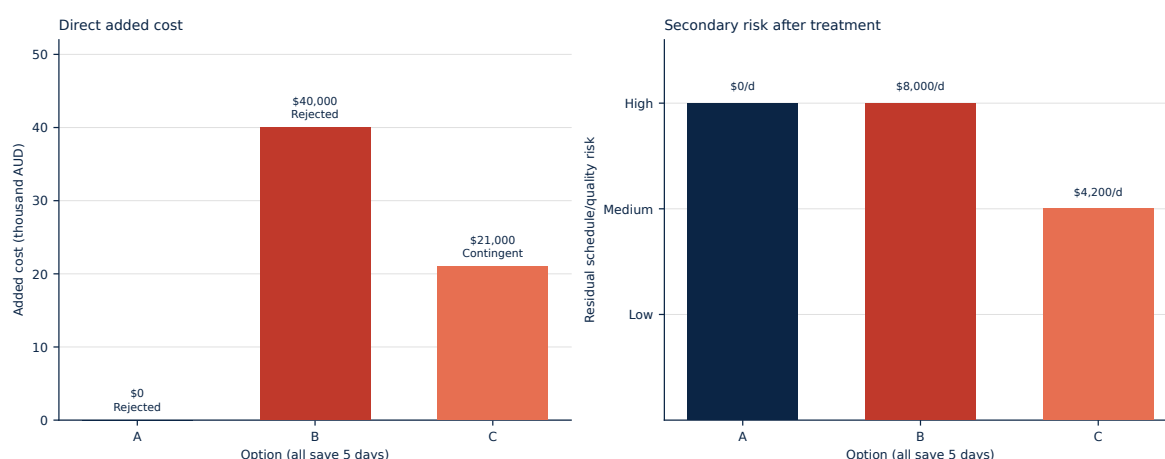


Figure 3.3: Compression trade-off for a five-day pull-in of $T-0$. Left: direct added cost. Right: residual schedule and quality risk after treatment. Option C is the only CPM-valid pull-in that keeps LRR at five days, countdown at four days, and pays $\$4,200$ per day rather than $\$8,000$.

4 Cost management

The cost baseline is the WBS in Figure 2.1 priced at Level 3. Seventeen control accounts sum to a work estimate of $\$1,500,000$. Contingency of $\$78,890$ is sized from residual expected monetary value plus four named residual items, not from a round-number markup, so the authorised budget at completion is $\$1,578,890$.

4.1 Estimating basis and assumptions

The estimate is built bottom-up at Level 3. Each package is $\text{hours} \times \text{an hourly rate}$, plus plant and materials. Four hourly rates are used.

A wage is not what an hour of work costs the project. The employer also pays super, leave and tax on every wage dollar. Some paid hours are never booked to a package. And a crew needs support that is not anyone's wage. The three multipliers below add those costs, one at a time, so each package pays its full share. All four rates start from the same wage and go through the same four steps.

1. **Wage.** The Manufacturing Award C10 rate is \$29.45 an hour from 1 July 2026 (Fair Work Commission, 2026). It is the legal minimum for a trade-qualified worker. Engineer and specialist rates start from the same wage so all four rates share one published base. C10 is not the award rate for those two jobs.
2. **Employment on-cost: what the employer pays on top of the wage.** Every wage dollar carries costs the worker never sees on their payslip. Each one is a share of the wage:
 - super 12% (Australian Taxation Office, 2025);
 - annual leave 7.69% (4 weeks off in 52, still paid) (Commonwealth of Australia, 2009);
 - leave loading 1.35% (17.5% extra on that annual leave, award clause 34.4) (Fair Work Commission, 2026);
 - personal leave 3.85% (10 days in 260) (Commonwealth of Australia, 2009);
 - public holidays 4.23% (about 11 weekdays in 260) (Commonwealth of Australia, 2009; Queensland Government, 1983);
 - long service leave 1.67% (8.6667 weeks after 10 years, so 0.867 weeks in 52) (Queensland Government, 2016);
 - Queensland payroll tax 4.75% (employers paying \$6.5 million or less in wages) (Queensland Revenue Office, 2024);
 - workers compensation 1.8% for office staff and 4.5% for pad staff (assumption).

WorkCover Queensland's average premium is 1.343% of wages in 2026–27, and each industry pays its own rate by risk (WorkCover Queensland, 2026). We assume 1.8% for office work and 4.5% for pad work, which sits well above that average. The first seven items add to 35.5%. With office workers compensation the total is 37.3%, so the multiplier is 1.373. With pad workers compensation the total is 40.0%, a multiplier of 1.400. For a technician, $\$29.45 \times 1.400 = \41.23 is the true cost of one paid hour.

3. **Utilisation (assumption): paid hours that cannot be charged.** Not every paid hour is spent on a package. Toolbox talks, safety briefings, waiting outside the blast zone and paperwork are paid but belong to no package. Those hours still have to be paid for, so they are spread over the hours that are charged. If 85 of every 100 paid hours are chargeable, each chargeable hour carries the cost of 100/85 paid hours, which is why we divide by 0.85. The share differs by role. Technicians are on the tools most of the day (85%). Engineers lose more time to reviews and reporting (72%). Specialists are on the pad only when their bars are live and wait in between (60%). These are our planning figures, not measured ones.
4. **Overhead (assumption): what the crew needs beyond wages.** A crew cannot work on wages alone. Supervision, site facilities, equipment and company running costs are paid by someone, and no single package owns them. The overhead multiplier adds that share to

every chargeable hour. It is 1.95 for technicians, 2.67 for engineers, 2.45 for specialists and 1.37 for surge. Surge is lowest because its wage has already been raised by the 1.5 overtime rate. These are our planning figures, set so the four charge-out rates sit where Table 4.1 shows them.

Table 4.1: Hourly charge-out build-up. The wage and statutory on-costs are published; workers compensation, utilisation and overhead are planning assumptions. Rates are rounded to the whole dollar.

Role	Wage	× on-cost	÷ utilisation	× overhead	Rate
Pad technician	29.45	41.23	48.51	94.59	\$95
Engineer / PM	29.45	40.43	56.15	149.92	\$150
Specialist	29.45	40.43	67.38	165.08	\$165
Surge technician [†]	44.18	61.85	72.76	99.68	\$100

[†]Surge is paid at overtime first: $\$29.45 \times 1.5 = \44.18 .

The crane is priced separately: a 100 t crawler at \$4,200 per wet day, operator included, from Queensland plant-hire rates checked against Rawlinsons (Rawlinsons, 2025).

The budget covers the seventeen packages in Figure 2.1. Factory manufacture, engine re-search and freight are out of scope, so A-100 has no cost. The surge crew (\$22,400) is already inside WBS 1.3.3. Option C's \$21,000 is not in the \$1,500,000; it would come out of the \$78,890 reserve only if R-14 happens.

Permits and Juru start on 10 August because they take too long to wait. Other August–September pad work is our choice. By Gate 1 (A-111, 2 October), \$602,301 of planned work is already under way: 40.2% of \$1,500,000. That is planned work, not cash paid. Gate 1 locks the other \$897,699 and the \$78,890 contingency rules. After that, adding pad work or moving *T-0* needs a Board variation.

4.2 WBS cost baseline

Table 4.2 is the rollup.

- Pad logistics (1.2): \$518,000 (34.5% of the work estimate).
- Pad testing (1.3): \$414,000 (27.6%).
- Governance (1.1): \$300,000 (20.0%).
- Flight/closeout (1.4): \$268,000 (17.9%).

Table 4.2: Bottom-up cost baseline by Level-3 WBS package. Totals equal the work estimate of \$1,500,000. A-100 is omitted because it is not a cost account.

WBS	Package	Amount (AUD)	Basis
1.1.1	PEP and integration	72,000	480 h PM @ \$150
1.1.2	Governance and EVM	48,000	320 h planner/cost @ \$150
1.1.3	ASA / CASA / GBRMPA	88,000	ASA \$42k + CASA \$28k + GBRMPA \$18k
1.1.4	Juru monitors and heritage	92,000	Two monitors + survey
1.2.1	Receiving and cleanroom	68,000	Technician hours + consumables
1.2.2	Crane and 23 m stacking	218,000	8 lift + 16 weld/NDT/bolt-up d × \$4,200 = \$100,800 + riggers
1.2.3	Cryo GSE	180,000	Valves/piping + specialist crew
1.2.4	RF array calibration	52,000	Crew + analyser hire
1.3.1	Avionics power-on	58,000	Engineers @ \$150/h + EGSE
1.3.2	WDR (cold)	160,000	Includes bulk LOX 25 t and LN2 \$68,500
1.3.3	Static fire	148,000	Includes surge hire \$22,400
1.3.4	Post-fire / pre-kitted spare	48,000	Engineers + spare handling
1.4.1	LRR and RSO	38,000	Review board; payload witnesses
1.4.2	Range exclusion / marine	58,000	Patrol boats + NOTAM
1.4.3	Launch and telemetry capture	82,000	Launch-day operations
1.4.4	Decrypt and handover	24,000	Five-day engineering handover
1.4.5	Demobilisation and audit	66,000	Site restore + audit
Base		1,500,000	

Notes. Labour rates are built in Table 4.1. Crane: 8 lift days and 16 weld/NDT/bolt-up days at the same wet \$4,200 (QLD 100 t crawler band about \$3,800–\$4,500/day), cross-checked against Rawlinsons (Rawlinsons, 2025). Weather dollars sit only in RES (4 extra wet days). 1.1.3: ASA 280 h @ \$150 = \$42,000; CASA CRIS airspace instrument \$28,000; GBRMPA permission \$18,000 (Australian Space Agency, 2025; Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019). LOX/LN2 \$68,500 is a bulk industrial-gas analogue inside 1.3.2. Surge \$22,400 = 4 × 7 d × 8 h × \$100 and sits in 1.3.3, not in contingency.

The largest package is crane mobilisation and stacking (1.2.2, \$218,000) because A-123 is 24 launch-critical days. Table 4.2 splits the crawler hire from rigger labour. The 16 non-lift days are plant time, not weather. Weather dollars sit only in RES: four extra wet days, \$16,800. R-06 is a dawn-lift control with EMV \$0, so it is not a second crane line. The seventeen lines sum to \$1,500,000. Package 1.2.1 stays one account for A-121 and A-122.

4.3 Contingency

Contingency is not a flat 15%. Table 4.3 adds residual expected monetary value to four named residual items. Treatments already sit in the \$1,500,000 base (surge hire, dawn-lift window, dual-path recorders, pre-kitted spare), so the pool uses residual $P'_{\%} \times I_{\$}$, not inherent $P_{\%}$. Ten cost-bearing risks sum to \$35,760. Itemised RES sums to \$43,130. Freight is out of scope and is not in RES. Together they are \$78,890 (5.26% of base), so *BAC* is \$1,578,890.

Table 4.3: Contingency build-up. The 5.26% ratio is residual EMV plus four named residual items, not an input markup.

Component	Amount (AUD)	Rule
Σ residual EMV of ten cost-bearing risks	35,760	Residual $P'_{\%} \times I_{\$}$; treatments already in the base
LOX/LN2 price variance (18% of \$68,500 gases)	12,330	In-scope commodity; not freight
Crane weather (only weather \$): 4 d $\times$ \$4,200	16,800	Extra wet days; R-06 is schedule-only
GN2/N2 purge: 40 G-size cylinders $\times$ \$200	8,000	Named multiply-out
Cryogenic kit: 3 $\times$ 3 m LOX hoses $\times$ \$1,400 + seal kit \$1,800	6,000	Named multiply-out
RES subtotal	43,130	Itemised; not a plug to 15%
Contingency reserve	78,890	5.26% of base
BAC	1,578,890	Base + contingency

Several register rows stay out of the \$35,760 so the money is not counted twice. R-08 (pad-civil defect) is transferred under the design-and-construct package and carries no Gilmour EMV. R-06, R-11, R-15, R-16, R-17 and R-18 are schedule or HSE exposures with no residual dollar draw. R-12 is the TestFlight 1 in-flight mode; the vehicle is factory product, so site-ops EMV is \$0. If R-14 is realised and Option C is invoked, the \$21,000 is drawn from the \$78,890 reserve, not added to the \$1,500,000 baseline. There is no management reserve above *BAC*: unknown unknowns would need a board variation this plan does not authorise.

4.4 Time-phased budget and cash flow

The performance measurement baseline spreads the work estimate on the Table 3.1 dates. Table 4.4 and Figure 4.1 use a uniform daily burn. October peaks because stacking, Wet Dress Rehearsal and static fire sit there. December is \$6,000 because only 1 December of A-145 falls in that month: $\$66,000/11 = \$6,000$. That is planned closeout, not unused contingency. Contingency is not monthly PV.

Table 4.4: Time-phased planned value of the \$1,500,000 work baseline. Contingency is not monthly spend.

Month	Period PV (AUD)	Cumulative PV (AUD)
Aug 2026	155,951	155,951
Sep 2026	407,188	563,139
Oct 2026 (peak: stack, WDR, static fire)	628,861	1,192,000
Nov 2026	302,000	1,494,000
Dec 2026	6,000	1,500,000

Figure 4.1 ends at \$1,500,000 and draws $BAC = \$1,578,890$ as an authorisation line, not December cash. Planned value is \$865,578 at M-4 (15 October) and \$602,301 at Gate 1. Option C's \$21,000 is outside Table 4.4. If spent, it is actual cost charged to contingency.

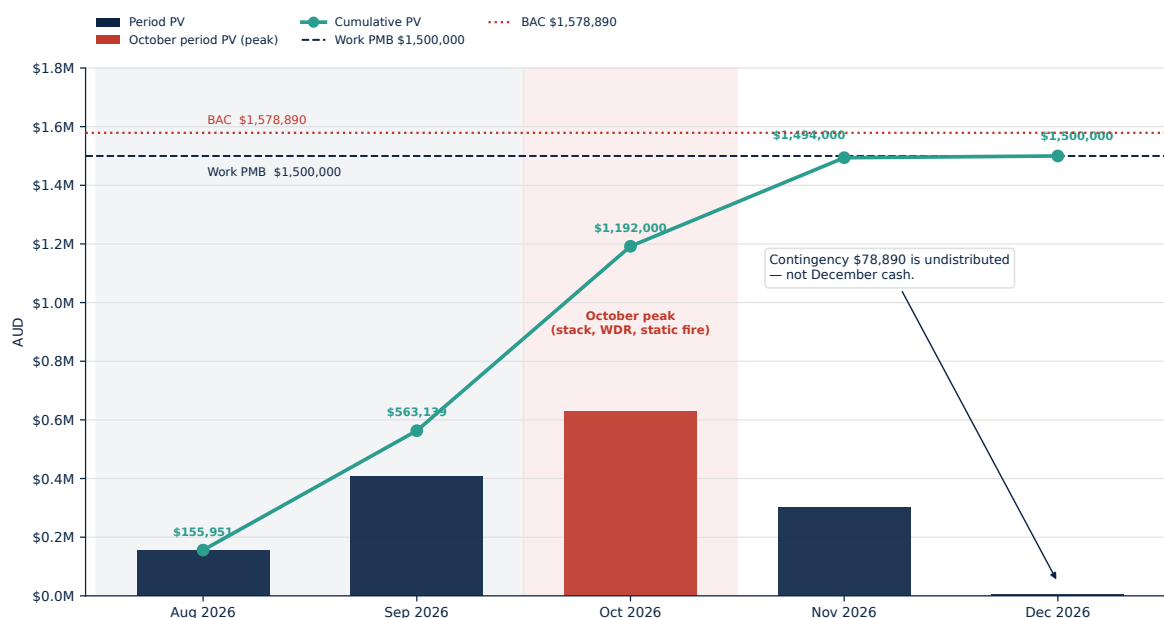


Figure 4.1: Cumulative planned-value S-curve for the Bowen campaign. Period bars are monthly PV; the teal line is cumulative PV and ends at the \$1,500,000 work baseline. The crimson dotted line is BAC (\$1,578,890) as an authorisation ceiling, not December cash. Contingency remains undistributed. Option C is not in the curve.

5 Resource management

This section names who owns each package, puts those people on the Table 3.1 dates, and shows where demand would pass twelve and what we do about it.

5.1 Roles and responsibilities

Table 5.1 is the RACI for this campaign. Responsible does the work. Accountable answers for it. Consulted sees the work before the Accountable owner signs. Each Level-3 package has exactly one Accountable owner.

There are two layers, both specific to Bowen.

Consultancy layer:

- Abdul Wahab Obeid (project manager and integration).
- Hadi Alkhub (planner).
- Ziyad Alahmari (risk, quality and HSE).
- Ilsa Hashmi (cost, resources and the Juru commercial interface).

Field layer:

- Launch Director.
- Range Safety Officer.
- GSE Lead.
- Avionics Specialist.
- Juru Cultural Officer.
- Design-and-construct package manager.

The owners follow the network.

- Hadi is Accountable for stacking because the dawn-lift window is a schedule control (R-06).
- The Launch Director is Accountable for static fire and lift-off, with the RSO consulted.
- Range exclusion is the other way round: the RSO is Accountable and the Launch Director is Responsible.
- Closeout audit is Ziyad's, so quality is not marking its own homework.

Table 5.1: RACI for Level-3 packages. Exactly one Accountable owner per row. Field titles are as used on the pad; consultancy names are the Group 1 officers.

WBS	Package	A	R	C
1.1.1	PEP and integration	Abdul	Hadi	Ziyad, Ilsa
1.1.2	Governance and EVM	Ilsa	Hadi	Abdul
1.1.3	ASA / CASA / GBRMPA	Abdul	Hadi	Launch Director
1.1.4	Juru heritage	Juru Officer	Ilsa	Abdul, Ziyad
1.2.1	Receiving / cleanroom	GSE Lead	D&C mgr	Hadi, Ilsa
1.2.2	Crane and stacking	Hadi	GSE Lead	Launch Director, RSO
1.2.3	Cryo GSE	D&C mgr	GSE Lead	Ziyad
1.2.4	RF calibration	Avionics	Hadi	Launch Director
1.3.1	Avionics power-on	Avionics	GSE Lead	Hadi
1.3.2	WDR (cold)	GSE Lead	Juru Officer	Ziyad, LD, RSO
1.3.3	Static fire	Launch Director	GSE Lead	RSO, Ziyad, Avionics
1.3.4	Post-fire / pre-kitted spare	Avionics	GSE Lead	Ziyad
1.4.1	LRR and RSO	Launch Director	RSO	Abdul, Ziyad, Juru Officer, payload sponsor
1.4.2	Range exclusion	RSO	Launch Director	Juru Officer
1.4.3	Launch (<i>T</i> -0)	Launch Director	RSO, Avionics	Abdul
1.4.4	Telemetry handover	Avionics	Hadi	Ilsa
1.4.5	Demob and audit	Ziyad	D&C mgr	Ilsa, Abdul

5.2 Resource allocation and the schedule

Every bar in Table 3.1 carries a named pad crew. The standing crew is eight: four GSE technicians, two stacking technicians, the Launch Director and the RSO. Juru monitors, avionics and RF specialists join when their bars are live. Eight is the standing organisation, not a hard pad limit.

The hard limit is twelve people on the pad at once, from the blast zone, suit air and access control. Figure 5.1 must not cross it. These occupancy numbers are PEP controls, not Week 4 charter figures. Demand comes from the schedule: each day, add the people on every live bar and count each person once. The weekly bar is that week's busiest day. Twelve is a same-time limit, not a weekly headcount.

5.3 Where demand peaks, and what we do

October is the peak because stacking, Wet Dress Rehearsal and static fire sit there. Leaving A-125 on its late dates (7–24 October) and adding four overtime heads on 16–30 October asks for fourteen people on 16–24 October: two over the cap of twelve, and six over the crew of eight. Week 9 is already thirteen, because late RF is still on the pad during stacking. After

24 October the RF specialists are gone, so leftover overtime sits on static fire. Figure 5.1 and Table 5.2 show that over-allocation. It is R-13.

Four responses were tested: smoothing, levelling, hiring and resequencing. The baseline is smoothing, then hiring. Levelling and resequencing are rejected.

Smoothing moves work only inside its float, so the finish date stays. A-125 has 39 days of total float and is not launch-critical. Placed on its early dates, 29 August to 15 September, the two RF specialists sit in weeks 3–6. Cost is \$0. Lift-off stays 15 November. Static fire on 25–30 October still needs more than eight standing heads. Wet Dress Rehearsal on 16–24 October is inside the cap once RF is gone, and it does not share a day with stacking.

Levelling would delay $TF = 0$ work. Wet Dress Rehearsal and static fire are already $TF = 0$, so pushing either pushes 15 November. Levelling is rejected.

Resequencing would start the rehearsal before stacking finishes, or static fire before permits and RF are done, and would break the hold points in Section 3. RF cannot follow static fire: the array has to be live before hot fire. Resequencing is rejected.

Hiring adds four surge technicians for seven days, 25–31 October, inside WBS 1.3.3 at \$100/h: $4 \times 7 \times 8 \times \$100 = \$22,400$, already in the \$1,500,000 estimate (Fair Work Commission, 2026). They replace the overtime heads. Mitigated 25–30 October is eight standing heads plus four surge: twelve, on the cap.

A ten-hour stagger on Wet Dress Rehearsal and static fire is the third R-13 control. Named people can share the calendar day without sharing the blast zone in the same hour. Cost is \$0 and no date moves. Figure 5.1 counts unique people that day, not the hour, so the figure does not change. Option C, if invoked, shortens post-fire diagnostics only and does not redraw it.

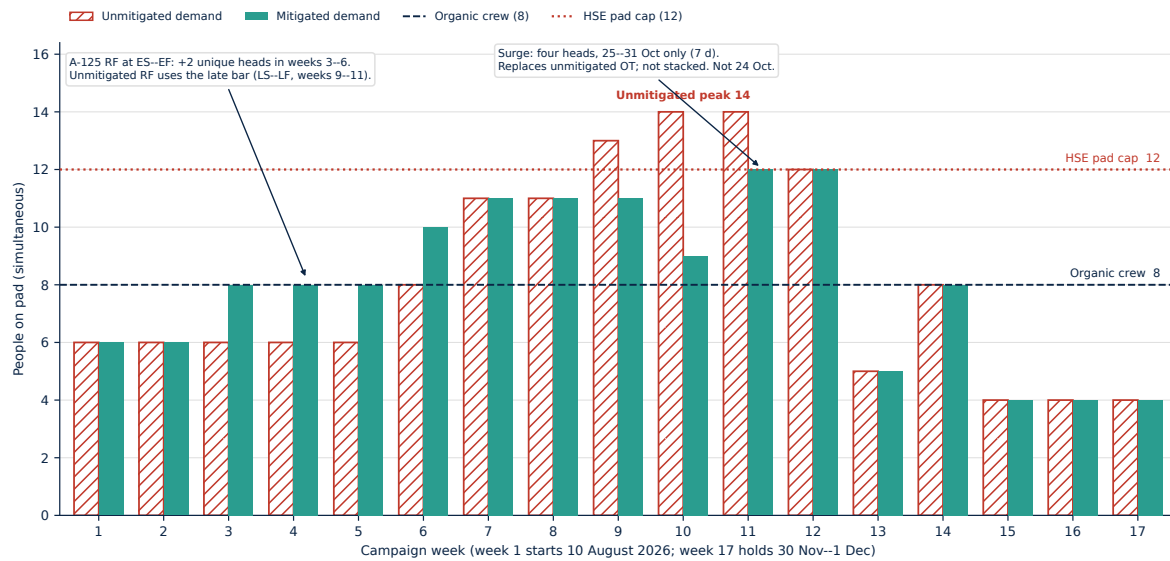


Figure 5.1: Weekly pad demand versus capacity, from Table 5.2. Each bar is the busiest day that week. Hashed crimson is the unmitigated series (peak 14 on 16–24 October): Wet Dress Rehearsal, RF still on its late bar, and four overtime heads. Teal is the mitigated baseline (peak 12): RF moved to weeks 3–6, and four surge technicians on 25–31 October only. The navy dashed line is the standing crew of eight. The crimson dotted line is the twelve-person pad cap. Eight is not a hard pad limit.

Table 5.2: Weekly simultaneous pad occupancy, from activity $\times$ named crew. Each weekly total is the maximum unique-head count on any calendar day that week (HSE cap is simultaneous, not weekly unique). Mitigated A-125 uses $ES-EF$; unmitigated uses the late bar $LS-LF$. Surge is four heads on 25–31 Oct only (7 days). Unmitigated overtime is four heads on WDR and static fire (16–30 Oct). Base, RF, surge and OT are disjoint name sets, so they add on the peak day.

Week	Mit. day	Base	RF	Surge	Mit.	Unmit. day	Base	RF	OT	Unmit.
1	10 Aug	6	0	0	6	10 Aug	6	0	0	6
2	17 Aug	6	0	0	6	17 Aug	6	0	0	6
3	29 Aug	6	2	0	8	24 Aug	6	0	0	6
4	31 Aug	6	2	0	8	31 Aug	6	0	0	6
5	7 Sep	6	2	0	8	7 Sep	6	0	0	6
6	14 Sep	8	2	0	10	14 Sep	8	0	0	8
7	22 Sep	11	0	0	11	22 Sep	11	0	0	11
8	28 Sep	11	0	0	11	28 Sep	11	0	0	11
9	5 Oct	11	0	0	11	7 Oct	11	2	0	13
10	12 Oct	9	0	0	9	16 Oct	8	2	4	14
11	25 Oct	8	0	4	12	19 Oct	8	2	4	14
12	26 Oct	8	0	4	12	26 Oct	8	0	4	12
13	6 Nov	5	0	0	5	6 Nov	5	0	0	5
14	15 Nov	8	0	0	8	15 Nov	8	0	0	8
15	21 Nov	4	0	0	4	21 Nov	4	0	0	4
16	23 Nov	4	0	0	4	23 Nov	4	0	0	4
17	30 Nov	4	0	0	4	30 Nov	4	0	0	4

5.4 Resource-conflict escalation

The live conflict is GSE technicians wanted on Wet Dress Rehearsal and on static fire in late October. The path is timed and named. It is not a values statement.

- Tier 1 is the field lead. They have four hours for a same-day fix: the ten-hour stagger, or borrow from float, with no extra spend. That is not calendar levelling. It does not delay a $TF = 0$ bar and it does not move 15 November.
- Tier 2 is the campaign PM (Abdul). They have twelve hours to choose overtime, hire or re-sequence if the stagger is not enough. Resequencing here is a live trade-off, not the baseline: the baseline already rejected changing the order of stacking, permits and hot fire. The cost goes to Ilsa so that Table 4.2 stays the book of record.
- Tier 3 is the Executive Board. They have 24 hours. They sit only if the choice would spend contingency, or would move static fire (M-5, 30 October) or lift-off (M-7, 15 November). Those dates are the same dates as Section 3.

6 Risk management

Campaign risk is owned by Ziyad Alahmari (Risk and Quality Lead). It is scored on this pad. The rows are not generic weather or budget-overrun lines. Table 6.1 has eighteen cause–event–effect statements. Those rows size the \$78,890 contingency already reported in Table 4.3. They also name the float that absorbs non-critical delay, and they fix the owners who treat each item.

6.1 Risk approach and scales

The process follows ISO 31000: identify, analyse, evaluate, treat, then monitor (International Organization for Standardization, 2018). Expected monetary value is a money sum, not a heat-map score. The 1–5 scores rank the rows. They are not dollars.

Inherent $EMV = P_{\%} \times I_{\$}$ uses the dollar column in Table 6.1. Contingency uses residual $EMV = P'_{\%} \times I_{\$}$. That is because the named treatments already sit in the \$1,500,000 base, so the reserve is only what is left after those treatments.

Probability bands:

- $P = 1$ below 5% (use 3% in EMV).
- $P = 2$ from 5–20% (12%).
- $P = 3$ from 21–40% (30%).
- $P = 4$ from 41–60% (50%).
- $P = 5$ above 60% (70%).

Schedule impact I_S :

- 1 (< 2 d).
- 2 (2–5 d).
- 3 (6–10 d).
- 4 (11–15 d).
- 5 (> 15 d on a $TF = 0$ path).

Cost impact I_C :

- 1 (< \$15k).
- 2 (\$15k–\$39,999).
- 3 (\$40k–\$79,999).
- 4 (\$80k–\$149,999).

- 5 (\$150k or more).

The matrix coordinate I is the higher of those two bands.

Residual P falls only where the named response actually bites. Residual I falls only where the effect itself is smaller. A 12–14 day recycle remains $I = 4$ even if it is less likely. Ziyad re-scores the register at Gate 1 (2 October), at M-4 (15 October) and at M-5 (30 October), in the same cadence as the EVM dashboard in Section 7.5.

6.2 Integrated risk register

Table 6.1 is the full set. Cause, event and effect are three columns. Owners match the RACI in Table 5.1.

- Ziyad sits with the GSE Lead on static-fire abort (R-01) and township acoustics (R-17).
- The Launch Director owns footprint, stage-separation and the in-flight TestFlight 1 mode (R-02, R-09, R-12).
- Ilsa Hashmi owns LOX logistics, occupancy and the Juru commercial interface (R-10, R-13, and R-04 with the Juru Cultural Officer).
- Hadi Alkhub owns the dawn-lift window and countdown software (R-06, R-18).
- Abdul Wahab Obeid owns permits and LRR deadlock (R-14, R-16).

Figure 6.1 plots every inherent point and the residual after treatment. R-02, R-09 and R-12 share high-impact cells and are offset.

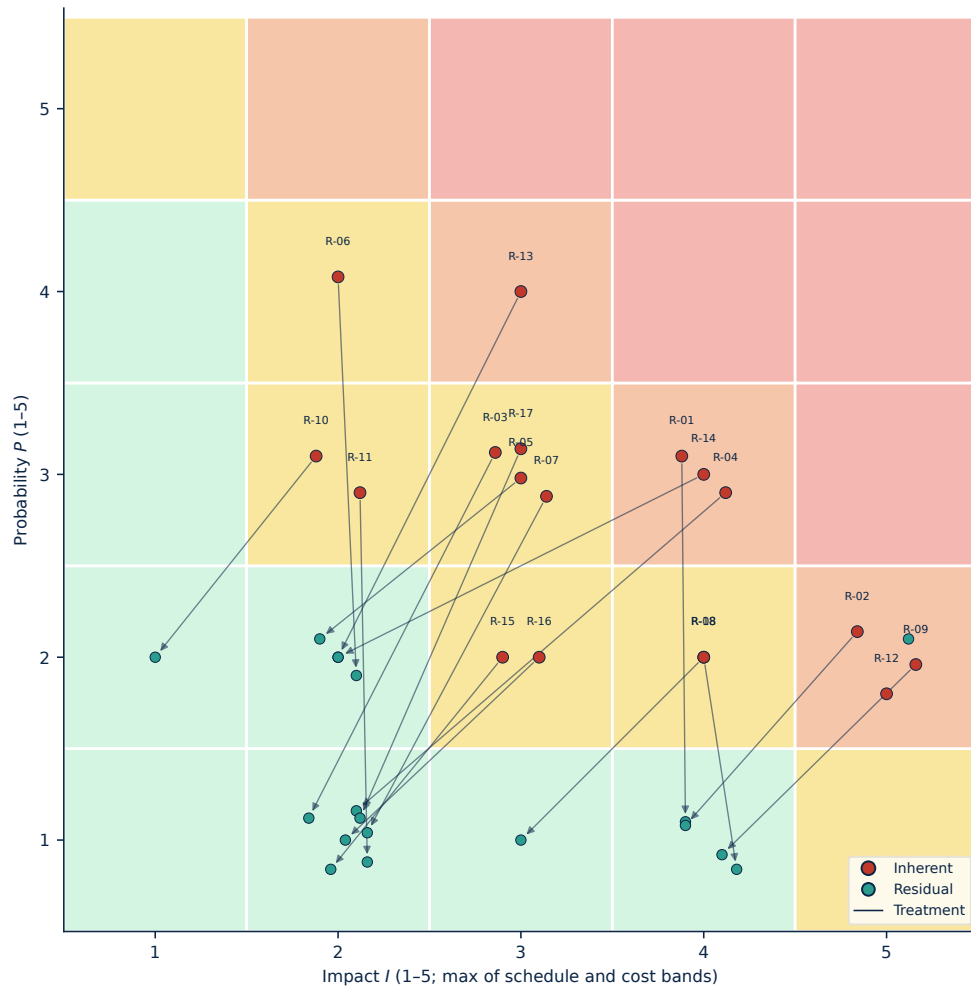


Figure 6.1: Inherent (crimson) to residual (teal) movement for all eighteen register items. P is vertical and I is horizontal. Most treatment arrows run down and left. R-01 keeps $I = 4$ (a recycle is still 12–14 days). R-08 keeps $I = 4$ (a hold-down failure is still a hot-ops stop; NDT drops P only). R-12 keeps $I = 5$ (in-flight loss still ends heritage). R-02, R-09 and R-12 share high-impact cells and are offset.

Table 6.1: Integrated campaign risk register. $P \times I$ uses the 1–5 scores. The EMV column is residual $P'_{\%} \times I_{\$}$, which feeds Table 4.3. A dash means the row is not in the \$35,760 pool.

ID	Anchor	Cause	Event	Effect	P	I	$P \times I$	Owner	Response	P'	I'	EMV
R-01	A-133 / 1.3.3	Residual pump insulation	Static-fire auto-abort	12–14 d recycle / \$90k	3	4	12	Ziyad / GSE Lead	Thermal soak; pre-kitted LRU spare; 48 h swap. Design defects return to R&D	1	4	2,700
R-02	A-143 / 1.4.3	Trajectory energy toward park boundary	Debris in GBRMPA zone	Statutory halt	2	5	10	Launch Director	South-biased LRR footprint; joint GBRMPA protocol	1	4	5,400
R-03	A-125 / A-143	RF interference at Bowen array	Telemetry <98.5%	Heritage fail	3	3	9	Avionics Lead	Dual-path recorders; A-125 before hot fire; mobile backup	1	2	1,500
R-04	A-114 / A-143	Exclusion without stop-work	Juru complaint	Range withheld	3	4	12	Ilsa / Juru Officer	Embedded monitors with stop-work; cultural HP before A-132	1	2	1,200
R-05	A-124 / A-132	Inadequate purge before LOX	Frozen/cracked valve	WDR abort ~8 d	3	3	9	GSE Lead	LN2 pre-chill; spare valve kit; helium leak test HP-2	2	2	6,600
R-06	A-123 / 1.2.2	Abbot Point crosswind >20 kt at vertical mate	Lift abort	Crane weather delay	4	2	8	Hadi / Stack Conductor	Dawn-lift window. Weather \$ only in RES (4 extra wet days)	2	2	—

Table 6.1: Integrated campaign risk register (continued).

ID	Anchor	Cause	Event	Effect	<i>P</i>	<i>I</i>	<i>P</i> × <i>I</i>	Owner	Response	<i>P'</i>	<i>I'</i>	EMV
R-07	A-124 / D&C	D&C GSE IFC lags propulsion ICD	Pad mismatch	Rework	3	3	9	GSE Lead / PM	ICD freeze gate; weekly integration; D&C LDs	1	2	2,100
R-08	1.2.3 / D&C	Latent pad-civil defect	Hold-down failure	Hot-ops stop	2	4	8	D&C mgr (A: PM)	NDT HP-1 drops <i>P</i> ; <i>I</i> stays 4 if a hold-down still fails; liability remains with D&C	1	4	0*
R-09	A-143	Stage-sep charge timing error	Debris outside footprint	Inquiry / halt	2	5	10	Launch Director	Dual-string sep inhibit; wind hold toward park	1	4	4,500
R-10	A-132 / 1.3.2	Single 800 km LOX tanker delay	WDR cannot load	2–5 d slip	3	2	6	Ilsa	72 h tanker confirm; second-tanker backup; RES on price	2	1	2,760
R-11	A-131	Flight-computer vs pad EGSE skew	Power-on abort	Avionics hold	3	2	6	Avionics Specialist	ICD freeze; cleanroom EGSE dry run before stack	1	2	—

Table 6.1: Integrated campaign risk register (continued).

ID	Anchor	Cause	Event	Effect	<i>P</i>	<i>I</i>	<i>P</i> × <i>I</i>	Owner	Response	<i>P'</i>	<i>I'</i>	EMV
R-12	A-143	Two-motor pump fault after <i>T</i> -0	Thrust loss as at T+14 s	Heritage fail	2	5	10	Launch Director	HP-3 screens ground abort; in-flight mode accepted (P stays 2); no on-pad redesign	2	5	—
R-13	A-132 / A-133	Peak 14 vs cap 12	Detank error	Recycle / injury	4	3	12	Ilsa	A-125 moved to Sep; four surge techs in base; 10 h stagger	2	2	3,600
R-14	A-113 / A-143	Late ASA filing or CASA NOTAM	15 Nov slot refused	Window missed	3	4	12	Abdul / Regulatory	File 60 d prior (A1 Keep Satisfied); weekly ASA/CASA liaison; Option C backup	2	2	5,400
R-15	A-142	Dugong in exclusion box	Sweep extension	Delay inside 4 d A-142 hold	2	3	6	RSO	Marine-mammal protocol; A-142 holds 4 d on the baseline and under Option C	1	2	—
R-16	A-141	RSO vs Launch Director deadlock at LRR	Missed 10 Nov gate	HP-4 unsigned	2	3	6	Abdul / Launch Director	Named DoA; escalate 24 h before A-141 <i>LF</i>	1	2	—

Table 6.1: Integrated campaign risk register (continued).

ID	Anchor	Cause	Event	Effect	P	I	$P \times I$	Owner	Response	P'	I'	EMV
R-17	A-133	Static-fire acoustic >135 dB at Bowen	Complaint / hold	Township stop	3	3	9	Ziyad / HSE	Water deluge; 1.5 km exclusion; community notice 14 d prior	1	2	—
R-18	A-142 / A-143	FTS/countdown abort at T -10	Recycle inside window	T -0 day consumed	2	4	8	Hadi / Avionics	Two countdown rehearsals; abort-to-recycle procedure	1	3	—

*R-08 is contract-transferred (EMV \$0 to Gilmour); residual I stays 4 because a hold-down that still occurs is still a hot-ops stop. R-06, R-11, R-12, R-15–R-18 are schedule, HSE or out-of-scope vehicle heritage; R-12 has no site-ops dollar draw. Residual EMV identities: R-01 $0.03 \times 90,000$; R-02 $0.03 \times 180,000$; R-03 $0.03 \times 50,000$; R-04 $0.03 \times 40,000$; R-05 $0.12 \times 55,000$; R-07 $0.03 \times 70,000$; R-09 $0.03 \times 150,000$; R-10 $0.12 \times 23,000$; R-13 $0.12 \times 30,000$; R-14 $0.12 \times 45,000$. Sum \$35,760.

6.3 Top risks, contingency, float and delivery model

Three rows dominate board attention because they sit on the launch-critical path and can stop $T-0$ or close the heritage case.

R-01 sits on static fire (A-133 / WBS 1.3.3). A residual pump-insulation defect can auto-abort the five-second fire: a 12–14 day recycle and about \$90k. The strategy is mitigate. Ziyad and the GSE Lead use thermal-soak instrumentation, a pre-kitted LRU spare and a 48-hour swap. Design defects still return to R&D. The score falls from (3, 4) to (1, 4): probability drops, and impact stays 4 because a recycle is still 12–14 days. Residual EMV is \$2,700 ($3\% \times \$90k$), not the inherent \$27,000, because the soak, spare and swap already sit in the baseline.

R-12 is the TestFlight 1 mode: thrust loss in two first-stage hybrid motors after $T-0$, as at $T+14$ s, which ends the heritage objective. The Launch Director owns HP-3 as a ground-abort screen. HP-3 does not treat an event after $T-0$, so P' stays 2 and impact stays 5. The strategy is accept: no on-pad redesign and no site-ops dollar draw (EMV \$0).

R-14 is an ASA flight authorisation or a CASA NOTAM refusing the 15 November slot. Abdul files 60 days prior, as the charter scored ASA Keep Satisfied, and holds a weekly ASA/CASA liaison. Crashing stacking does not treat R-14: permits already finish on 24 October with $TF = 0$. The strategy is escalate. Option C goes to the Board if this row fires. No 10–12 November payload geometry is used.

R-09 (stage-separation timing) and R-02 (footprint toward the park) remain high-impact register rows on the same LRR protocol. They are not the three board headlines.

Those ten rows are the \$35,760 residual EMV in Table 4.3. The \$43,130 RES covers LOX price variance, crane weather (the only weather dollars), GN2/N2 purge stock and a cryogenic hose/seal kit. Together they are \$78,890, 5.26% of base, not a 15% markup. Freight stays out of RES. R-06 has EMV \$0, so weather is not counted twice.

Float, not money, is the treatment for several feeders.

- A-125's 39 days let Ilsa park RF in weeks 3–6 so R-13 never meets an occupancy of 14.
- A-124's three days absorb GSE delay before Wet Dress Rehearsal (A-132) without moving 15 November.
- A-114's five days absorb heritage delay before Wet Dress Rehearsal (A-132) without moving 15 November.

Delivery risk uses the model in Section 7.2. An ICD lag (R-07) is a liquidated-damages event on the design-and-construct contractor. A latent pad-civil defect (R-08) stays with that package after HP-1 NDT. The strategy is transfer: residual EMV is \$0 to Gilmour, and residual I stays 4 because a hold-down that still fails is still a hot-ops stop. Abdul stays Accountable and the package manager stays Responsible.

Option C is a secondary risk. Shortening post-fire diagnostics to one day creates extra labour and quality exposure on that bar. Section 3.3 prices that labour at \$21,000 and keeps the thermal

and electrical screens. If R-14 fires, Abdul takes Option C to the Board and Ilsa draws \$21,000 from contingency. The draw is not in the \$1,500,000 baseline, Figure 3.2 does not move until the Board says so, and A-142 stays a four-day marine hold.

7 Delivery and control

This section turns the baselines in Sections 3–6 into decision rights, procurement, hold points, Bowen HSE, and the earned-value framework read at M-4. The campaign has not been executed.

7.1 Governance and reporting

Governance is the organisation that will stop work. The Gilmour Executive Board is the sponsor. Abdul Wahab Obeid is Campaign Project Manager. Figure 7.1 and Table 5.1 name the consultancy leads and the field titles.

Delegation is numeric.

- Abdul may approve a variance of at most \$20,000 *and* at most 48 hours, and only if M-5 (30 October, A-133) and M-7 (15 November, A-143) do not move.
- Anything else—a contingency draw, Option C’s \$21,000, a hold-point waiver, a *T*-0 change—goes to the Board within 24 hours.

R-16 (RSO versus Launch Director deadlock) escalates 24 hours before A-141’s late finish of 10 Nov, not inside the review room. The \$20,000 ceiling is below the dollar *impact* of the hot-fire rows that would recycle the pad (R-01 \$90,000; R-07 \$70,000). Residual EMVs on those rows are \$2,700 and \$2,100 because treatments already sit in the work baseline; a genuine recycle still cannot be buried in PM discretion.

Cadence matches the physical work.

- Field leads hold a daily pad standup.
- After Gate 1 (A-111, 2 Oct) Ilsa publishes a weekly EVM dashboard from A-112.
- Abdul takes a fortnightly pack to the sponsor.

Daily standup does not rewrite Table 3.1. A-112 has 53 days of float so the control system does not compete with stacking for pad occupancy.

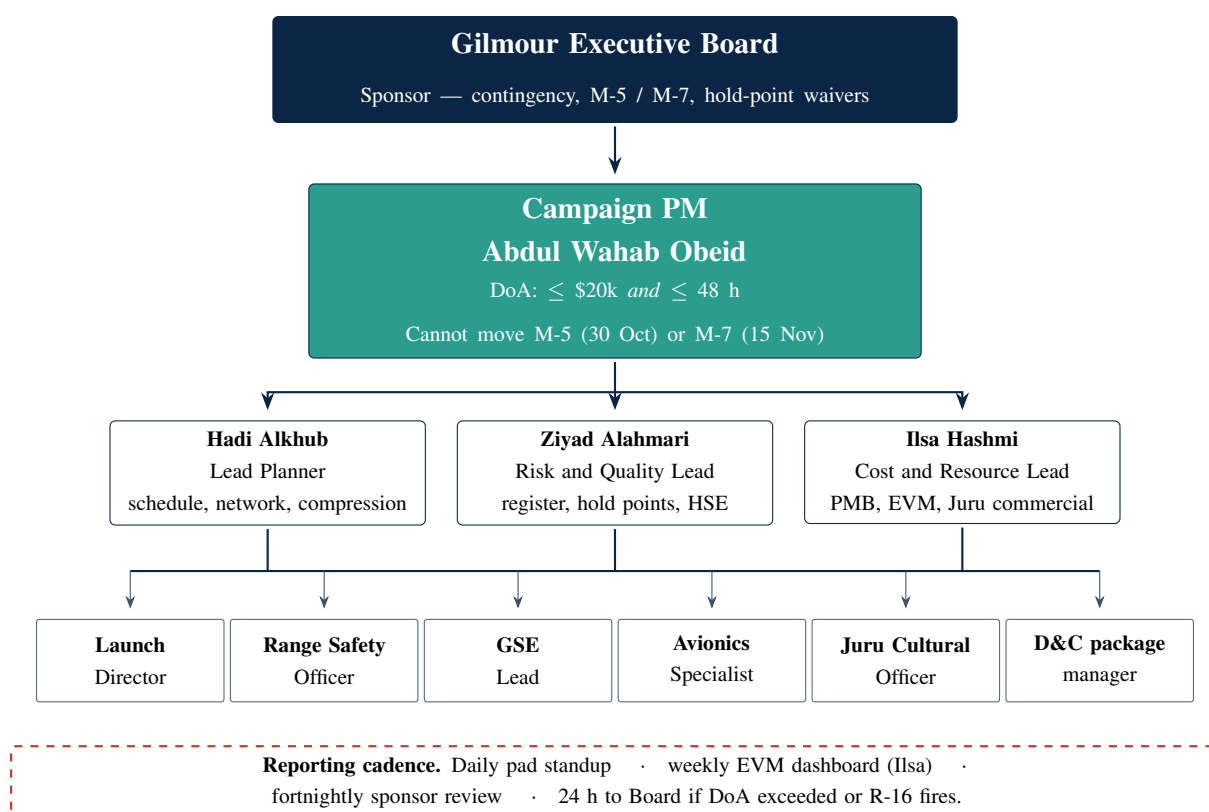


Figure 7.1: Campaign governance for Bowen pad operations. Named roles, a numeric DoA on the PM, and the reporting cadence. Field titles match Table 5.1. The Board, not the PM, moves M-5 or M-7 or draws the \$78,890 reserve.

7.2 Delivery model and procurement

The chosen delivery model is a managing-contractor / design-and-construct hybrid. Gilmour retains propulsion intellectual property, vehicle integration and the Australian launch licence. Civil works and cryogenic GSE packages are let as design-and-construct, with an interface-control freeze and liquidated damages. That split is the contractual answer to R-07 and R-08: an ICD lag is an LD event on the contractor, and a latent pad-civil defect stays with the D&C package after HP-1 NDT (R-08 EMV \$0 to Gilmour). Abdul stays Accountable on that row; the package manager stays Responsible.

Turnkey EPC was rejected. After TestFlight 1, giving the propulsion stack and the launch authorisation to one EPC would put flight-critical intellectual property and statutory accountability in a contractor who does not hold the licence (Gilmour Space Technologies, 2025; Australian Space Agency, 2025). Two tests decide the model. Scope certainty is high on site operations (seventeen Level-3 packages, locked dates, a \$1,500,000 work estimate) and low on the vehicle. Client capability is the other test: Gilmour can integrate a launch vehicle; pouring pad grout or running a 100 t crawler is not its core work. The charter set the weights at 40/30/30, and A1 did not score the options. This PEP scores the site-ops choice only:

- Strategic Fit 40% (EPC 2/5, D&C 5/5).

- Technical Feasibility 30% (3/5 vs 4/5).
- Schedule 30% (3/5 vs 4/5).

Weighted 2.6 versus 4.4, so D&C wins. Alliance contracting was the second rejected alternative: a four-month campaign cannot carry pain-share overhead, and GBRMPA and CASA duties are not commercially shareable (Great Barrier Reef Marine Park Authority, 2019; Civil Aviation Safety Authority, 2024).

7.3 Quality

Quality uses two standards together (SAE International, 2016; International Organization for Standardization, 2015). ISO 9001:2015 is the general rule: plan the work, name who checks it, and do not pass a job that has not been verified. AS9100D adds hold points, extra checks on welds, leaks and ignition, and a formal release before flight. This campaign uses those rules as pad stops.

ISO 9001 maps onto named owners. The PEP is the quality plan and Ziyad is the release authority. Table 5.1 names who is Accountable on each package. Design defects found on the pad go back to R&D. If HP-1 is unsigned, stacking earns \$0.

AS9100D maps onto the four hold points in Table 7.1. A hold point is a stop: the next job cannot start until the check is done and the hold is released. HP-1 is weld NDT on A-123 (R-08), HP-2 the helium leak test on A-124 (R-05), HP-3 the ignition interlock on A-133 (R-01), and HP-4 LRR sign-off on A-141. The payload sponsor witnesses HP-4 and does not sign. Work does not pass a hold without Ziyad's release or a Board waiver, which sits outside Abdul's authority. Option A was rejected because signing HP-4 on incomplete post-fire data would be a hold-point failure.

Table 7.1: Quality hold points for the Bowen campaign. Each hold is a named artefact, not a meeting. Measurement rules in Section 7.5 follow the same IDs.

Hold	Activity	Risk	Artefact	Release
HP-1	A-123 (15 Oct)	R-08	Pad tie-down weld NDT	D&C mgr / Ziyad
HP-2	A-124 (12 Oct)	R-05	GSE helium leak test	GSE Lead / Ziyad
HP-3	A-133 (30 Oct)	R-01	Pre-static-fire ignition interlock	GSE Lead / Launch Director / Ziyad
HP-4	A-141 (10 Nov)	R-16	LRR sign-off (payload sponsor witnesses)	Launch Director / RSO / Juru Officer / Abdul / Ziyad

- Discrete tests and flight artefacts—A-132, A-133, A-141, A-143—are 0/100.
- A-113 is elapsed-day percent-complete because the work is a 76-day wait, not a binary test.

- Installations—A-123 and A-124—are milestone percent-complete tied to HP-1 and HP-2.

LRR waits on A-134 rather than on A-133: HP-4 cannot be signed on incomplete post-fire data, which is why Option A was rejected (SAE International, 2016).

7.4 Health, safety and environment

HSE is this pad under the *Work Health and Safety Act 2011* (Qld) (Queensland Government, 2011).

- Liquid oxygen is the process hazard: LOX PPE, oxygen-clean tools, and exclusion during load and detank on A-132 and A-133.
- This PEP's pad controls—not A1 charter numbers—are a 1.5 km blast zone and twelve simultaneous heads, the same cap on Figure 5.1.
- Acoustic treatment is a water deluge so static-fire level at Bowen township stays at or below 135 dB (R-17), with community notice fourteen days prior.
- Juru artefact buffers and stop-work authority (R-04) are HSE controls: A-114 feeds A-132 as a cultural hold.
- Finish-line HSE remains as chartered: zero lost-time injuries, zero GBRMPA breaches, and zero Juru heritage breaches.
- Marine-mammal protocol on A-142 (R-15) is owned by the RSO (Great Barrier Reef Marine Park Authority, 2019).
- If R-14 fires, Option C shortens A-134 only and keeps A-141 at five days and A-142 at four days so HP-4 and the marine hold are not crashed.
- Occupancy never exceeds twelve on the mitigated histogram: R-13 is already treated in the \$1,500,000 base by moving A-125 and hiring four surge technicians.

7.5 Earned value management

Earned value tells the Board whether the \$1,578,890 authorisation is still intact (Project Management Institute, 2019). It is a framework. The campaign has not started, so there is no EV/AC history to plot.

Distributed planned value is \$1,500,000, the same total as Table 4.2 and Figure 4.1. Contingency of \$78,890 is undistributed: not December cash and not monthly PV. Authorised *BAC* is \$1,578,890. Option C's \$21,000 sits outside the work baseline; if the Board invokes it, the draw is actual cost charged to contingency.

SV and CV are currency, never days:

$$SV = EV - PV, \quad CV = EV - AC, \quad SPI = EV/PV, \quad CPI = EV/AC. \quad (5)$$

In-flight estimate at completion on the work baseline is

$$EAC = AC + (BAC_{\text{work}} - EV)/CPI, \quad (6)$$

with $BAC_{\text{work}} = \$1,500,000$. The same CPI is applied to the authorisation ceiling as $BAC \times AC/EV$, reported against \$1,578,890 (Project Management Institute, 2019). Finish-line rule: final $AC \leq \$1,578,890$. A Green CPI of 0.96 in October is allowed; finishing above BAC is not.

Thresholds that trigger action, used only in this section, are:

- **Green** — $0.95 \leq CPI, SPI \leq 1.05$: dashboard only.
- **Amber** — $0.90 \leq \text{index} < 0.95$: Exception Report and a seven-day recovery plan within 48 hours.
- **Red** — $\text{index} < 0.90$, or $T-0$ slip > 2 days: contingency freeze and Board intervention. SV is not written in days.

The only dashboard numbers in this PEP are the two illustrative readings at M-4 (15 October 2026) in Table 7.2 and Figure 7.2. Planned value at that date is \$865,578, with HP-1 signed (A-123 100%), HP-2 signed, and A-113 on elapsed-day percent-complete. On-plan books EV equal to PV (\$865,578) and AC \$901,000, so $CPI = 865,578/901,000 = 0.96$ and $SPI = 865,578/865,578 = 1.00$ —Green. Even so,

$$EAC = 901,000 \times 1,500,000/865,578 = \$1,561,384,$$

which matches $AC + (BAC_{\text{work}} - EV)/(EV/AC)$. Authorisation

$$BAC \times AC/EV = 1,578,890 \times 901,000/865,578 = \$1,643,503$$

sits above the \$1,578,890 ceiling. Green at M-4 still carries a recovery obligation toward the finish-line rule.

The late-stack case holds PV at \$865,578 but drops EV to \$647,578 and AC to \$870,000 because HP-1 is unsigned. A-123 is milestone 0/100 on that hold, so the \$218,000 account earns \$0 of EV . Then $CPI = 647,578/870,000 = 0.74$ and $SPI = 647,578/865,578 = 0.75$ —Red. Ilsa issues the Exception Report within 48 hours and freezes contingency. If R-14 has fired, the seven-day recovery invokes Option C and crashes A-134 only, not A-141 or A-142. Work EAC is $870,000 \times 1,500,000/647,578 = \$2,015,201$. $BAC \times AC/EV = 1,578,890 \times$

$870,000/647,578 = \$2,121,187$. These two points are one worked example at M-4, not a claim that October has already been worked.

Table 7.2: Illustrative earned-value status at M-4 (15 October 2026). The campaign has not been executed. PV is the locked work-PMB value at stacking complete. Indices use the reported two-decimal CPI/SPI from EV/AC and EV/PV .

Status at M-4	PV	EV	AC	CPI	SPI	Action
HP-1 signed (A-123	865,578	865,578	901,000	0.96	1.00	Green — weekly dashboard
100Illustrative on-plan						
Illustrative late stack	865,578	647,578	870,000	0.74	0.75	Red — 48 h exception report; contingency freeze

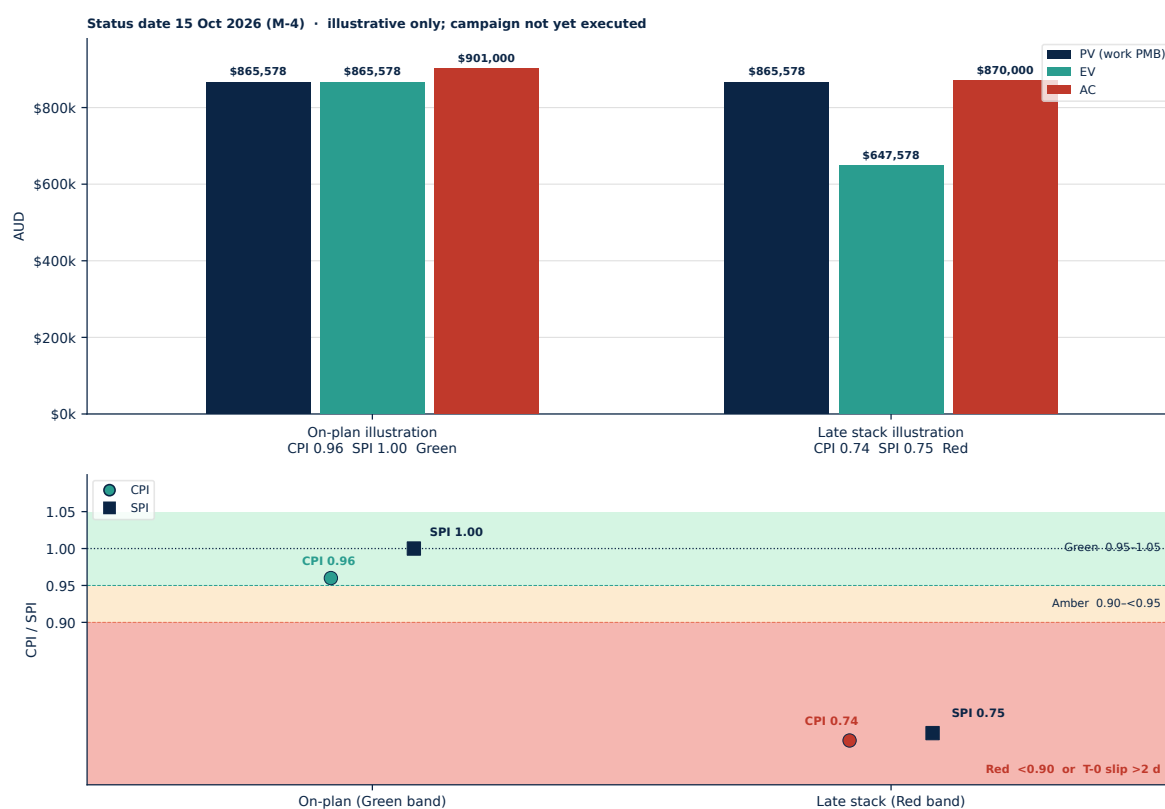


Figure 7.2: Illustrative status at M-4; campaign not yet executed. Upper panel: PV, EV and AC at the single status date 15 October 2026 for the on-plan and late-stack cases. Lower panel: Green / Amber / Red CPI–SPI bands, with the on-plan pair (0.96, 1.00) in Green and the late-stack pair (0.74, 0.75) in Red. Red is also triggered by $T-0$ slip greater than 2 days. No weekly EV/AC history is implied.

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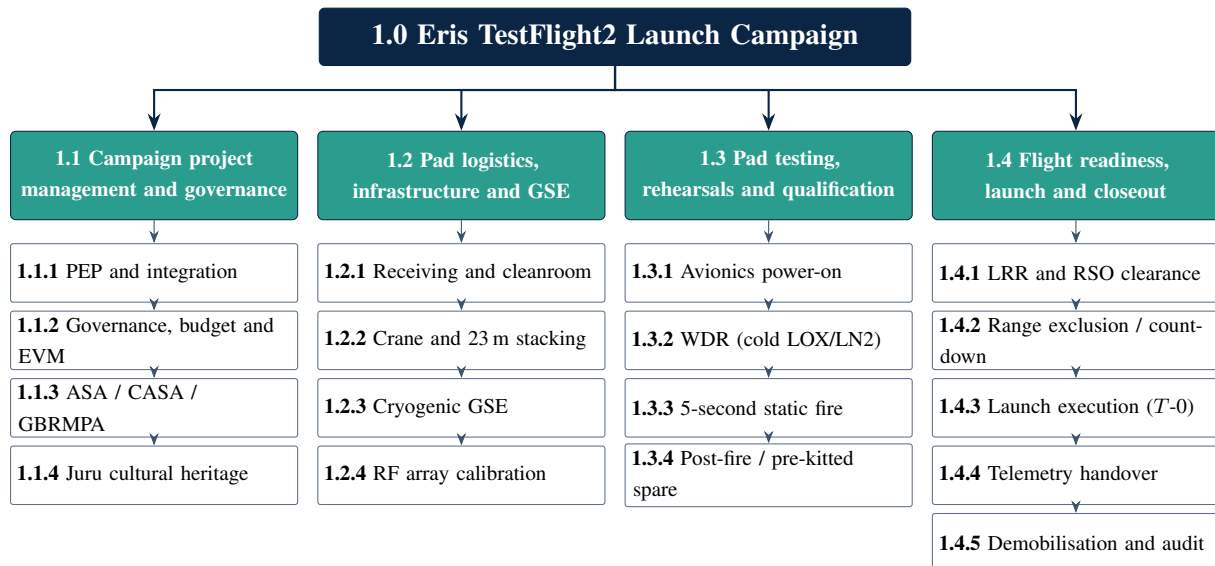
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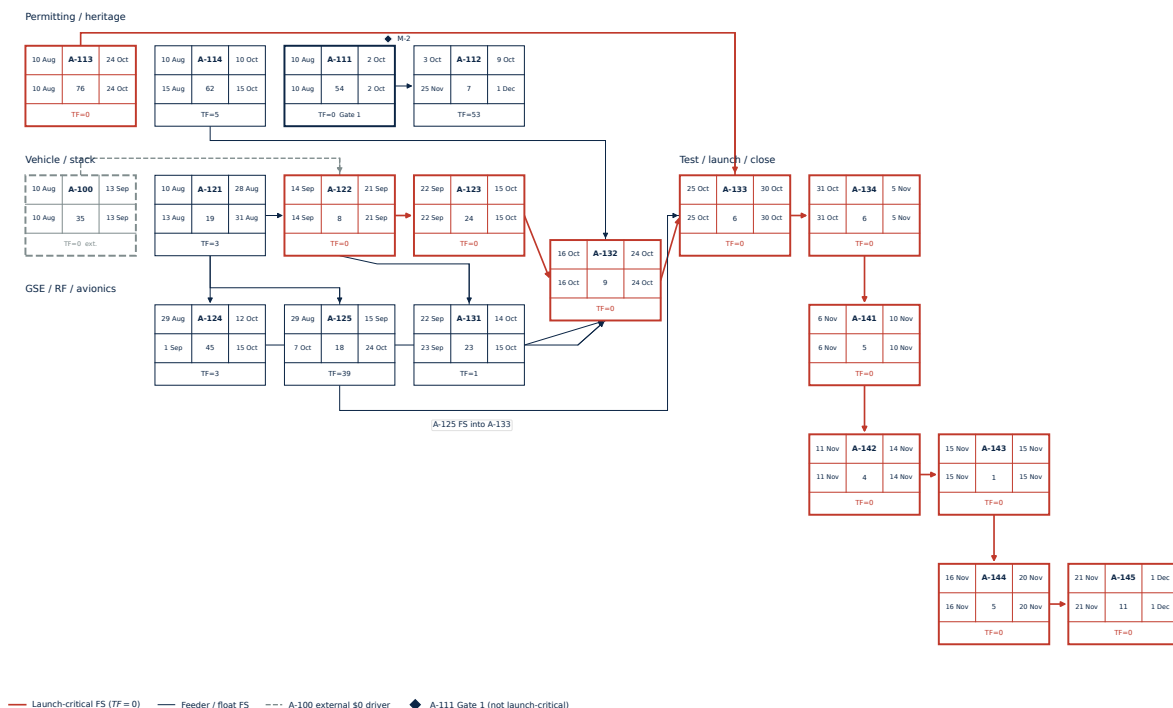
Appendix A: Campaign figures

This appendix reprints every figure used in the PEP. The originals stay in their sections and keep their original numbers. Nothing here is a new figure.

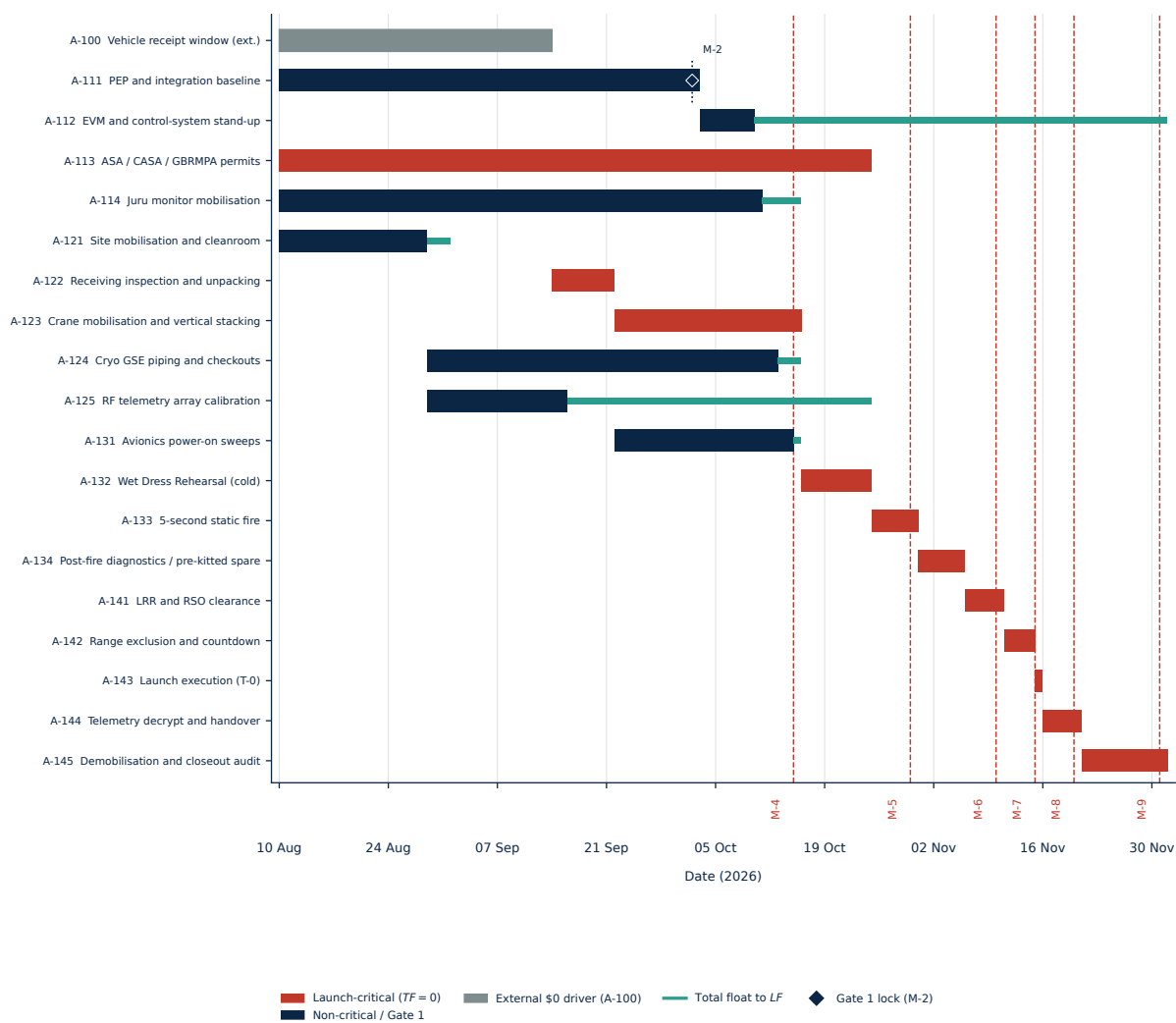


EXT A-100 — not a WBS cost account. External \$0 freight driver into A-122; omitted from Table 4.2.

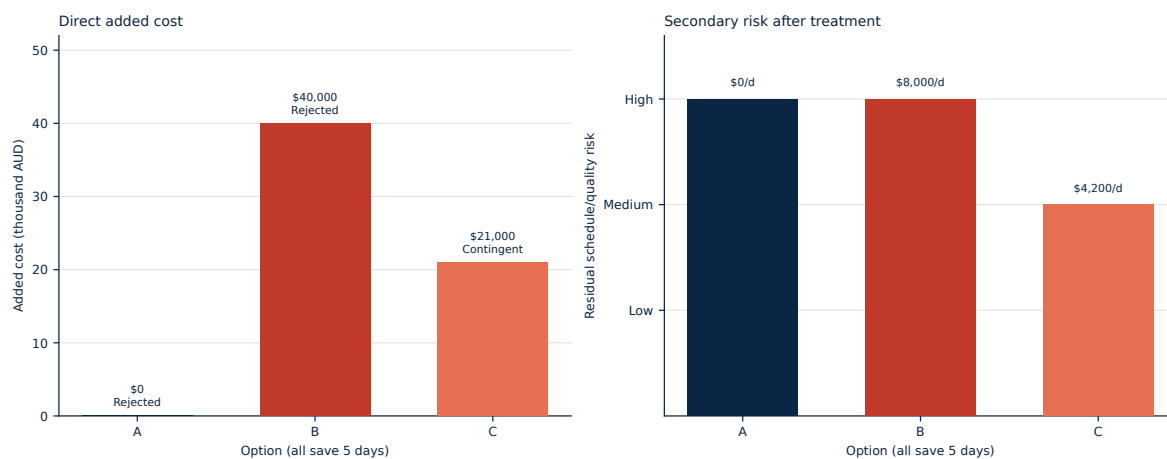
Repeat of Figure 2.1. Work breakdown structure for Eris TestFlight2 launch-site operations.



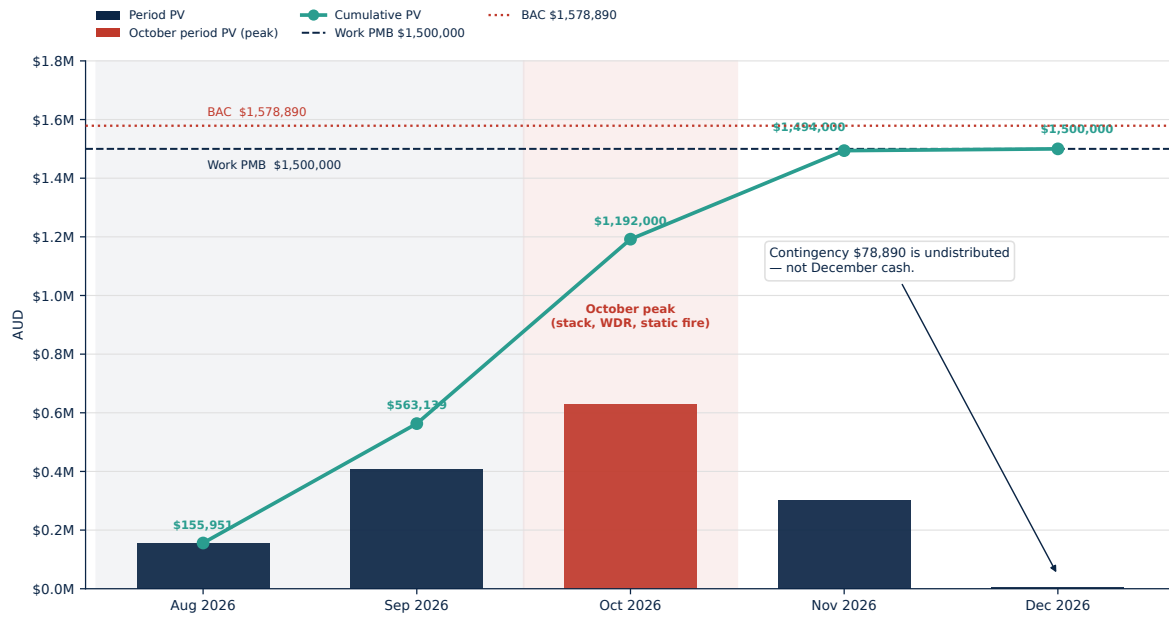
Repeat of Figure 3.1. Activity-on-node network for the Bowen campaign.



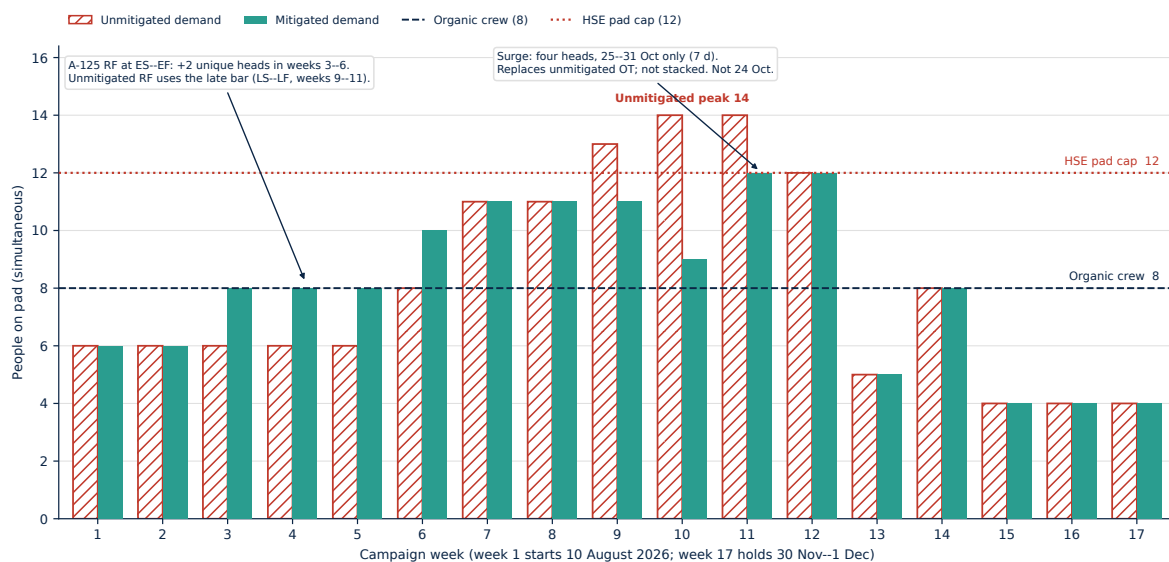
Repeat of Figure 3.2. Baseline Gantt from 10 August to 1 December 2026.



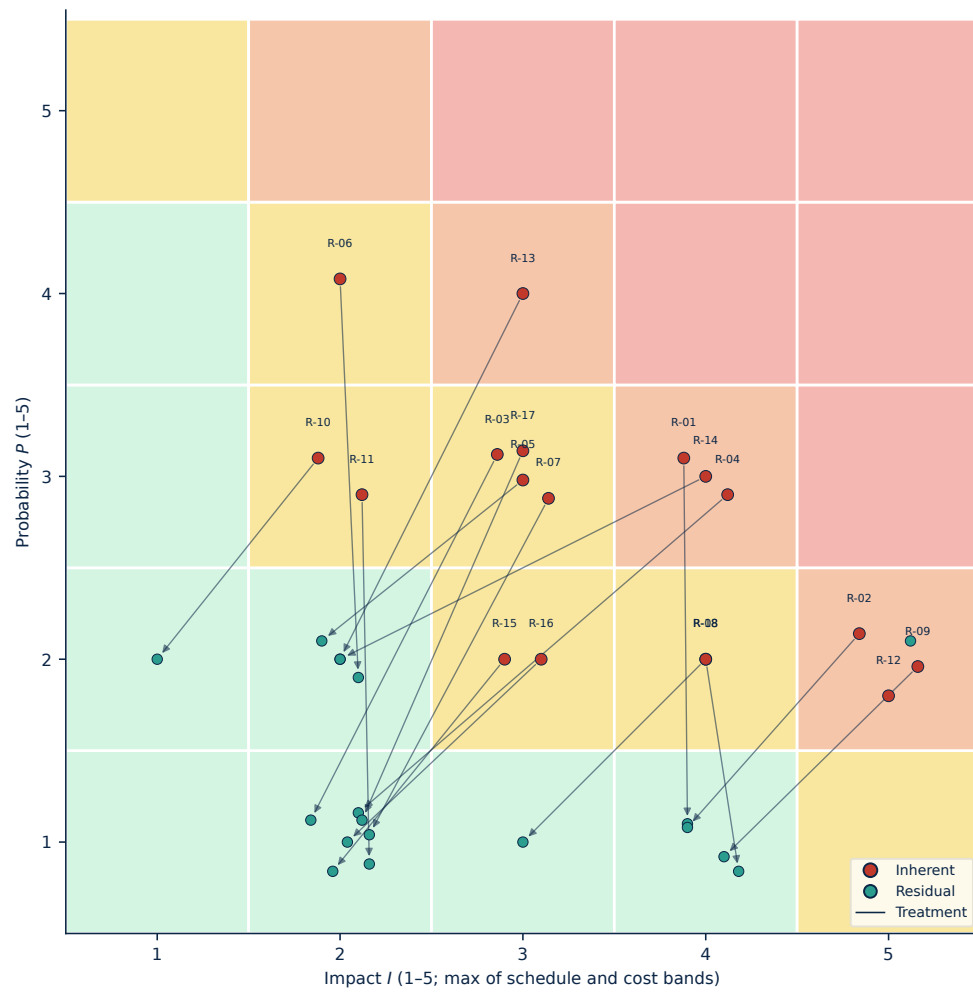
Repeat of Figure 3.3. Compression cost and risk trade-off.



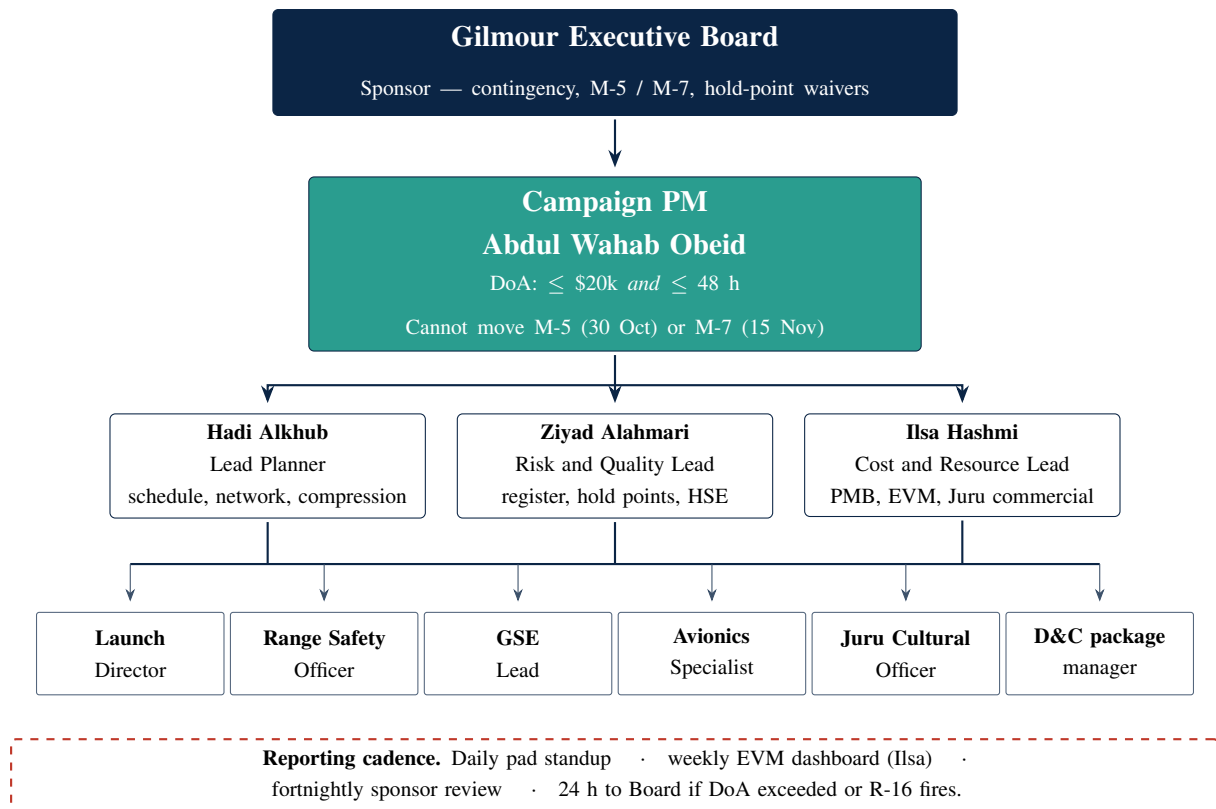
Repeat of Figure 4.1. Planned-value S-curve.



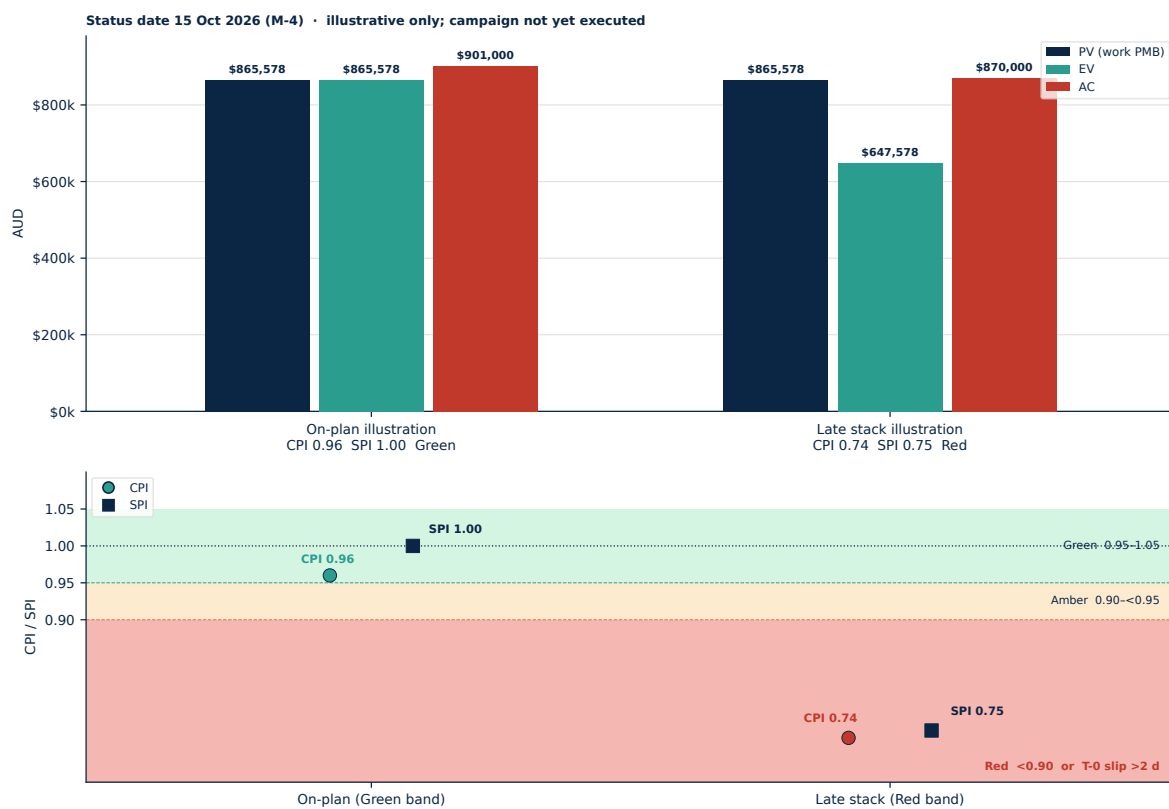
Repeat of Figure 5.1. Weekly pad demand versus capacity.



Repeat of Figure 6.1. Inherent-to-residual risk matrix.



Repeat of Figure 7.1. Campaign governance structure.



Repeat of Figure 7.2. Illustrative earned-value chart at M-4.